Inverse Earth Studio: a reproducible geophysical workbench for processing, forward modelling, inversion, and scientific review

Felipe Santibáñez-Leal

Technical report · v0.04.003, 13 November 2026 · Licence CC BY 4.0
CAOS Research open-problems programme · product Inverse Earth Studio · geophysics
Version DOI: not assigned
Concept DOI (always latest): not assigned
Not peer reviewed. Source code, data and computational records:
github.com/fsantibanezleal/CAOS_Geophysics

Abstract

Geophysical inversion is underdetermined: observations constrain a subsurface model only through acquisition geometry, a forward operator, error assumptions, parameterization, prior information, and numerical optimization. Inverse Earth Studio is an open-source research workbench connecting data ingestion and processing, forward modelling, inversion, diagnostics, and interpretation across potential fields, electromagnetics, electrical methods, and seismology. Its implementation includes Python pipelines and solvers, bilingual computational notebooks, provenance-bearing experiments, browser instruments, and a bounded authenticated-service design.

The method suite spans gravity and magnetic processing and inversion, magnetotelluric screening and layered-earth inversion, electrical-resistivity quality control and inversion, waveform processing and phase picking, travel-time inversion, acoustic full-waveform inversion, joint inversion, and learned estimators. Results include both recoveries and failures. For example, a synthetic gravity case has weighted WRMS 1 while recovered model correlation is 0.283 and centroid depth is about 570 m for a 282 m truth; a low-velocity FWI case remains unresolved. Clear Lake station cl061 is retained for tensor quality control and excluded from 1D inversion after failing the consistency screen. The geometry-normalized travel-time refinement study is a negative synthetic result: on the joint family/acquisition holdout, two same-seed CUDA runs have velocity RMSE ratios of 2.194 and 2.190, and independent travel-time RMSE ratios of 5.585 and 5.595 relative to the selected weighted-L2 reference. Every joint-shift case exceeds its run-specific validation threshold. These are fixed synthetic cohorts, not field-performance estimates or independent-training-seed evidence. Null-space projection is treated as established prior methodology, not as a novel contribution of this product.

The workbench makes data fit, model recovery, and geological interpretation separate questions. Synthetic cases carry known truth; field observations do not. Replayed browser artifacts are distinguished from online calculations, and notebook execution is distinguished from scientific validation. The report documents the methods, numerical assumptions, data provenance, executable results, and limitations; it does not claim that unresolved scientific or service-acceptance gates have passed.

1 Introduction

An inverse image can look geologically persuasive even when multiple models explain the data, the assumed physics are violated, or a withheld acquisition subset is poorly predicted. This is familiar across geophysics: potential-field inverses are non-unique; MT dimensionality and component quality condition whether a 1D interpretation is defensible; seismic inversion depends on illumination, bandwidth, starting model, and cycle skipping; and coupled or learned estimates inherit their structural or training priors. A low objective is evidence about a specified objective under a specified data model. By itself, it is not evidence of unique or accurate geology.

The topic sequence draws on the public theoretical-practical geophysical inversion course by Mantilla [13]. Its notebooks, recordings, images, and raw data are not redistributed. The source commit was `0084021fcc6c1d2f112b144e6e5f0a3d92f20dff` (Mantilla course). The research survey distinguishes primary sources from implementation choices and identifies where the methods differ from cited work (docs/research/review.md). The aim is to connect domain assumptions, executable calculations, diagnostic evidence, and failure states within one inspectable project.

The report covers gravity station correction and prism inversion, aeromagnetic processing, induced-magnetization modelling, magnetotelluric screening and layered-earth inversion, electrical resistivity inversion, response-aware waveform processing, first-arrival tomography, acoustic full-waveform inversion, joint potential-field inversion, learned seismic-velocity estimation, and earthquake phase picking. It also evaluates a proposed coverage-conditioned null-space correction. Method definitions and numerical assumptions are in docs/design/SDD.md.

The intended reader moves between physical interpretation and software inspection: a geophysicist can ask what an observation constrains, while a developer can follow the same sample through parsing, processing, forward prediction, inversion, evaluation, and export. The workbench is organized around that chain rather than a single image of a buried object. Gravity, magnetics, and electromagnetic methods have different depth sensitivities and units; refraction and waveform methods depend on travel paths, bandwidth, and source-receiver geometry. Their shared presentation therefore emphasizes method-specific variables, physical axes, masks, residuals, and status labels instead of a common visual score. Each case must explain whether it is synthetic or measured, what evidence is available, and what conclusions remain unsupported.

The product's public course is inspired by a cited geophysical inversion course, but the source notebooks, recordings, and datasets are not copied into this repository. The English and Spanish lessons are authored to teach the implemented methods and their own data contracts. Their executable status and measured depth are independent of learning effectiveness, which has not been evaluated in a controlled study. Likewise, the breadth of the method catalogue does not mean that every advertised physical workflow accepts arbitrary user observations or produces a field-ready model. The report records the evidence now available and the tests still needed to make those workflows complete.

2 System architecture and evidence design

2.1 Evidence chain

The intended data flow follows the path

```
source → immutable raw asset → validated dataset → processing → model/survey →
```

`solve → evaluation → result/export`.

At each boundary the contracts carry modality, named dimensions and order, coordinate frame or specified local coordinates, units, geometry, masks and reasons, uncertainty, parent identities, code/environment information, and hashes as applicable. Unknown units, datum, signs, component orientation, rights, or response conventions are reasons to reject or quarantine an input rather than infer missing facts. Known truth is attached only to explicitly synthetic cases. Field observations may be screened and processed, but a synthetic constructor is never substituted for missing field provenance or truth.

The application separates three execution contexts. Visitors can inspect public material and use only those browser calculations whose client-side behavior has been validated. Authenticated project workflows are designed to isolate upload, ownership, and bounded worker execution on the single canonical ML VPS origin. Larger training, extensive FWI, and full bake workflows remain local CPU/GPU pipelines. A static or browser replay of a saved artifact is not represented as an online inverse solve. The specified deployment uses one bounded ML VPS service. Its capacity and numerical job behavior have not been measured on the live host.

2.2 Reproducible software artifacts

The software contains a Python data pipeline and solver implementations, generated and source-controlled scientific outputs, JSON contracts and validation outputs, frontend TypeScript/React instruments, and computational notebooks in English and Spanish. The user-facing browser is intended to display computed arrays and explicit physical axes, not to invent a second approximate result for animation. Physical result records distinguish observations, predictions, signed residuals, model values, objective terms, coverage and uncertainty diagnostics, stopping state, and provenance. Display interpolation or color mapping is a rendering operation and must not be mistaken for increased model resolution.

Notebooks are part of the research material, not merely screenshots. Their lessons explain assumptions, execute the named calculations, show numerical outputs, compare with independent identities or baselines when available, and retain unresolved results. The bilingual content was corrected where translations, counts, or numerical interpretations differed. Execution and translation checks establish neither field validity nor educational efficacy.

3 Problem formulation

For observations $\mathbf{d} \in \mathbb{R}^N$ and a discretized model $\mathbf{m} \in \mathbb{R}^M$, the forward problem evaluates a method- specific operator $\mathbf{F}$ under a declared acquisition, coordinate frame, and physical approximation. The inverse problem estimates model parameters consistent with the data and an explicit error model while using a prior to control directions the observations do not determine:

$$\mathbf{d} = \mathbf{F}(\mathbf{m}) + \boldsymbol{\epsilon}, \qquad \widehat{\mathbf{m}} = \arg\min_{\mathbf{m}\in\mathcal{C}} \frac{1}{2}\|\mathbf{W}_d(\mathbf{F}(\mathbf{m}) - \mathbf{d})\|_2^2 + \beta\,\mathcal{R}(\mathbf{m} - \mathbf{m}_{\text{ref}}). \tag{1}$$

The error vector $\boldsymbol{\epsilon}$, weight $\mathbf{W}_d$, admissible set $\mathcal{C}$, regularizer $\mathcal{R}$, trade-off β, and reference $\mathbf{m}_{\text{ref}}$ define the estimator as much as the solver. If errors are independent and correctly scaled, diagonal weights may use inverse standard deviations; correlated errors require an appropriate covariance model. Observation units, coordinate reference, geometry,

masks, and uncertainties therefore belong to the numerical problem, not to display metadata. A residual score with no named transformation and units is not comparable across physical methods.

For a linearized potential-field operator $\mathbf{G}$, an invisible model perturbation obeys

$$\mathbf{G}\,\delta\mathbf{m} = \mathbf{0}. \tag{2}$$

Regularization selects an estimate among models that fit similarly; it does not add observations to the null space. Mesh spacing, survey placement, depth weighting, bounds, and the reference model can all change the selected structure. For a nonlinear forward model, the local Jacobian $\mathbf{J}(\mathbf{m})$ describes local sensitivity and does not prove global uniqueness. An optimizer's stopping state or a small training loss is a numerical fact. Recovery is a separate conclusion that needs a predeclared metric and, for synthetic tests, known truth.

The method families instantiate different physics and parameterizations. Gravity and magnetic inverses use volume sensitivities and density or susceptibility variables. Layered MT uses complex transfer functions over frequency. ERT depends on electrode geometry and conductivity. First-arrival times integrate slowness over paths, while waveform inversion solves a wave equation and fits time samples. Joint objectives introduce a structural or petrophysical coupling assumption; learned estimators add a training distribution and representation. Their objective values are not a common quality scale. A readable result must name its data, operator, prior, weights, constraints, stopping rule, and evaluation variable. When a field input lacks source, units, datum, orientation, or error metadata needed by that method, the correct state is ineligible or QC-only, not a model with guessed conventions. The objective must also be interpreted with its units and geometry intact. A normalized residual may help balance scales but can change noise variance; a regularizer may stabilize an estimate but can bias the recovered amplitude or shape. Two optimizers applied to different parameterizations or weights do not constitute a controlled algorithm comparison. This is why each comparison fixes a target, operator, starting model, acquisition, and selection protocol before comparing test outcomes. A visually smooth model can be a consequence of its prior, and a low data misfit can result from fitting noise or an incorrect forward model. Independent physical checks and held-out prediction address different failure modes and are reported separately.

4 Notation

The cross-method frame uses x for easting, y for northing, and z for depth positive downward unless the original format explicitly defines otherwise. A coordinate value is not assumed to be a geographic datum. The vectors $\mathbf{d}$, $\mathbf{m}$, and ϵ denote observations, model parameters, and errors only within the relevant method; their dimensions and physical units vary. The symbols table is a reading aid and does not imply that density, conductivity, magnetization, velocity, and slowness share a numerical scale. Each artifact's axis labels and units remain authoritative for interpreting its values.

Symbol	Meaning	Unit or convention
N, M	Observation and model dimensions	counts
$\mathbf{d}$, $\mathbf{F}(\mathbf{m})$	Observed and predicted data	method-specific
$\boldsymbol{\epsilon}$	Data and modelling error	data units
$\mathbf{W}_d$	Data-weighting operator	inverse data units
$\mathcal{R}$, β	Regularizer and trade-off	method-specific
$\mathbf{m}_{\text{ref}}$	Reference or starting model	model units
$\mathbf{G}$, $\mathbf{J}$	Linear operator and local Jacobian	method-specific
σ, Σ	Marginal scale and covariance	units and squared units
f, t, ω	Frequency, time, angular frequency	Hz, s, rad/s

Subscripts are local to an experiment: i commonly indexes observations, j model cells, k frequencies or time samples, and s sources or shots. Superscript T denotes transpose; a complex conjugate transpose is written H. In MT, $Z_{ab}(f)$ is a complex transfer function whose units, axes, and tensor convention come from the source record. In waveform processing, $u(t)$ denotes recorded amplitude and $p_P(t)$ denotes a model probability; they are distinct quantities. In travel-time methods, t_i is an observed or predicted arrival according to context, and $s = 1/v$ denotes slowness when the section explicitly uses it.

Units are converted only at a named processing boundary. A magnetic field in nanotesla must not be compared directly with a vector stored in tesla without its scale conversion. A log-resistivity parameter is not itself measured in ohm-metres, although exponentiating it may produce a physical resistivity. Coverage, sensitivity, and resolution also have different meanings: ray counts describe support, Jacobian columns describe local response, and resolution requires an explicit operator or experiment. None is a standard deviation unless a calibration procedure establishes that interpretation. Method sections define additional symbols at first use and carry grid ordering and axis orientation into their artifact contracts. When arrays are reshaped, the declared order is part of the numerical definition: for example, a depth-major velocity tensor and a northing-major surface map cannot be displayed with the same implicit transpose. Sampling interval and acquisition time are distinct from the physical coordinate vector, and radians per second are distinct from hertz by a factor of 2π. Complex-valued transfer functions retain their real and imaginary parts together; amplitude and phase plots are derived views. A missing value is represented by a mask and reason rather than a numeric sentinel that could be mistaken for a measurement. These conventions make exported files interpretable outside the browser and reduce the chance that a display transform is silently treated as a scientific operation.

5 Data and data contracts

3.1 Authored synthetic experiments

The original canonical suite uses independently constructed geological geometries instead of repeating one centered anomaly. Its earlier 0.04 report describes 20 distinct cases and six controlled conditions per case (120 experiments, 348 method results). The conditions vary contrast, noise, acquisition, coverage, or regularization according to the physical family; they are not interchangeable perturbations and are not samples from a geological population. Later repository work adds separate survey, uploaded-data, learning, and source-screening experiments. Therefore the older 120-experiment suite is a defined artifact family, not a count of every experiment now present in the repository.

Synthetic constructors provide known truth for controlled tests of forward physics and

conditional inversion behavior. They do not encode the full complexity of field geology. Noise perturbations at fixed mesh/prior/tradeoff measure conditional estimator repeatability, not posterior uncertainty. A held-out line, shot, family, or acquisition tests only the specified split and comparator. Repeated numerical trials from one generator do not establish field performance.

3.2 External data and eligibility

Two SimPEG tutorial archives are independently pinned and locally preprocessed as ingestion examples. The source documentation states that their derived observation tables are not redistributed while upstream redistribution rights remain unresolved. They are not field validation. The measured Clear Lake EDI station cl061 is hash-pinned and cited to the USGS and EarthScope data releases [15, 16]. Its impedance tensor fails the required 1D consistency screen; it remains a QC-only dataset, and no layered inversion or geological result is claimed.

Other data paths include provider-linked or rights-restricted data. For example, the Slagdump ERT raw asset is absent locally under its provider-link-only status; focused notebook checks consequently skip tests that require those bytes. The Koenigsee travel-time lesson is an authored synthetic method control and does not establish execution on the provider-linked case. The STEAD excerpts [6] and associated project-trained checkpoint have a scoped redistribution decision with attribution; that decision does not generalize to other data or establish model accuracy. Rights, data custody, field eligibility, numerical correctness, and scientific validation remain separate decisions.

5.1 Data contracts

Experiment catalogue

The experiment catalogue is the public index of baked examples. Each case identifies its physical family, geometry, seed, and available variants. A variant points to an artifact with an expected path, byte count, and SHA-256 digest, and describes the methods, metrics, status, target quantity, units, and target provenance that the interface may display. This declaration allows a client to locate a reproducible array and verify that the served bytes match the committed inventory. The catalogue is not an inverse solver, an uncertainty model, or a scientific acceptance certificate. A row can correctly report an unresolved or failed result. The product guard checks artifact closure and a separate numerical validation checks selected arrays and calculations; neither replaces interpretation of the stated target and evaluation criteria. The type and versioned schema are recorded with the catalogue contract.

Data readiness

The readiness record answers whether the declared product artifacts exist and agree with their expected inventory. It carries the product and catalogue versions and provides a deterministic status for a release candidate. Readiness is intentionally narrower than scientific quality: a complete byte inventory can contain a failed experiment, QC-only observation, or synthetic result with limited transfer. It does not clear source rights, validate units or datum, establish that a field survey is eligible for an inverse, or prove that a solver recovered geology. Downstream interfaces must keep this meaning visible and must not upgrade a readiness state into a scientific verdict. The readiness schema and independent

validator provide the boundary; the catalogue's method evaluations and method-specific evidence remain separate. This division allows a release to be reproducible while still recording unresolved science without hiding it.

Private API

The API contract separates open public exploration from protected project operations. Public viewers can inspect documentation and use browser calculations whose client-side behavior is specified. Project creation, ownership, uploaded data, saved processing jobs, and server-side computation require authenticated access. Requests and responses carry declared ownership and lifecycle states so the UI can distinguish an unavailable service, a rejected input, a queued operation, and a completed result. The OpenAPI document is the route-level source for this boundary, while the runtime authorization and state checks remain implementation responsibilities. The API is currently not accepted for production use: schema presence and local tests do not prove public TLS behavior, resource limits, cancellation, isolation, backup, or deployment admission. The approved test system has one canonical ML-VPS origin; there is no second Pages origin.

API payloads

The payload schemas describe data submitted to and returned from the private workflows. They bind a payload to its modality, version, source or parent identity, units and coordinate metadata where known, processing parameters, and hashes for the raw or scientific values. Method-specific shapes retain arrays and masks instead of flattening their physical meaning: for example, EDI impedance keeps its complex tensor and aligned frequency axis, while waveform records retain sample time and channel metadata. Eligibility reasons and QC outcomes are represented explicitly so that malformed, incomplete, or unsupported measurements are not silently converted into inverse results. Current schemas cover the declared API surface, but nested constraints are not yet exhaustive for every advertised modality. A schema-valid payload therefore still needs parser, physics, rights, ownership, and method-specific validation before a calculation is accepted.

Browser model assets

Browser-executable learned assets use a public manifest that binds the model file, dimensions, normalization, preprocessing, and evaluation records to exact digests. The browser runtime must consume the declared checkpoint and arrays; it must not silently substitute a differently trained model or recompute a second approximate dataset. This contract supports reproducible asset loading and parity comparisons for the specified CPU and WebAssembly paths. The recorded phase-picker parity uses 23 selected QC-valid display traces from the same checkpoint; its maximum output discrepancy and peak displacement are runtime-consistency measures only. They do not establish agreement with an independent training framework, population accuracy, field transfer, or analyst acceptance. Model training and full inverse solves remain offline workflows. A valid asset manifest proves file identity and metadata shape, not predictive quality or a novel method claim.

6 Methods

The method identifiers below follow docs/design/SDD.md. Each entry names an implementation and its tests; this does not imply validated use for every input or a complete end-to-end workflow. Table 1 summarizes the implemented purpose and evidence limits, while Figure 1 shows one source-bound, synthetic processing example.

Table 1: Methods, executable purpose and evidence limits across the implemented geophysical method set. Rows summarize current implementations and bounded datasets; they do not imply field-wide validity.

Method	Executable purpose	Principal limitations
Gravity corrections and transforms	Gravity station corrections, datum and uncertainty lineage, and transforms with safeguards against duplicate corrections.	Field eligibility still depends on exact source, datum, instrument, and uncertainty lineage. A local adapter pass does not validate arbitrary field tables.
Weighted 3D gravity prism inversion	Weighted 3D gravity prism forward/inverse calculations, including a bounded L2 survey inversion and comparison with independent prism physics.	Non-uniqueness persists; a good data fit or a visually compact body does not identify geology. Full-survey and field performance remain untested.
Aeromagnetic processing and continuation	Aeromagnetic line/tie-line/height checks, blocked line holdout, support-aware gridding, and physically supported upward continuation.	Authored survey controls do not establish broad provider-data validation or field magnetic inversion.
Induced-magnetization prism modelling	Induced-magnetization prism forward modelling and physical objective/derivative construction, checked against Choclo.	A full nonzero survey inverse with held-out evaluation is not included in the available evidence. Remanence violates the induced-direction assumption.
EDI quality and dimensionality screening	MT EDI component, variance, rotation, tipper, and dimensionality screening.	A tensor record is not automatically eligible for a 1D inverse. cl061 remains QC-only.
Layered-earth impedance inversion	Layered-earth complex-impedance response and bounded 1D inversion, with cross-checks against MTpy and browser/local analytic cases.	Known layer thickness and isotropic 1D assumptions constrain the synthetic inverse. The cl061 field screen fails.
Two-dimensional electrical resistivity inversion	2D ERT processing/inversion with electrode geometry, topography, reciprocal/QC logic, blocked holdout, and explicit failed-inverse states.	Rights-restricted raw data are not fabricated or represented as executed; independent comparison and some field paths remain open.
Response-aware waveform processing	MiniSEED/StationXML waveform handling, response-aware filtering, STA/LTA or matched-arrival picks.	Response epochs, units, gaps, and channel orientation are strict eligibility conditions. STA/LTA picks are not universal truth labels.
First-arrival travel-time inversion	First-arrival travel-time inversion with explicit source/receiver geometry, bounded velocity, ray coverage, and held-out-shot comparison against a baseline.	The executed teaching control is synthetic and uses a straight-ray/Dijkstra model; the synthetic control does not represent provider-linked field data.

Method	Executable purpose	Principal limitations
Acoustic full-waveform inversion	Acoustic constant-density FWI with Deepwave, bounded velocity updates, full-band and frequency-continuation schedules, active/withheld receiver diagnostics, and saved optimizer states.	Non-convexity, cycle skipping, limited illumination, inverse crime from using the same forward engine, and acoustic-only physics remain material limitations. Some cases improve model RMSE but fail the specified noise-consistency criterion.
Joint potential-field inversion	Joint potential-field inversion comparing uncoupled fits, spatial cross-gradient coupling, and an explicit Gaussian-mixture petrophysical prior.	The project Gaussian-mixture-prior implementation is not the SimPEG PGI optimizer. Coupling can trade one property against another; one calibration matrix does not establish general performance.
Learned seismic-velocity estimation	A learned seismic-velocity inversion trained on disjoint procedural families and compared with a same-input weighted L2 solve and independent quadrature forward operator.	The frozen joint family/acquisition holdout fails both prespecified ratios; training success does not override the held-out result.
PhaseNet-family phase picking	PhaseNet-family P/S inference [7], matched STA/LTA comparison, and local browser ONNX inference with explicit preprocessing and abstention rules.	Curated browser parity is runtime consistency, not accuracy. Population metrics come from the separate locked test cohort and analyst references, not geological truth.
Geometry-normalized travel-time residual refinement	Backprojected travel-time residuals are normalized by ray length and cell coverage, then supplied with the coverage field to a bounded residual CNN.	The current physics-v2 study fails both predeclared joint-shift ratios in the primary and same-seed replay. This is a negative synthetic comparison; null-space projection is established prior methodology and no superiority claim is made.

6.1 Method-specific mathematical formulations

Gravity station corrections and uncertainty

The station workflow treats a gravity value as a measurement with a history, not as an unlabeled scalar. Each row retains the source value, station identifier, coordinate reference, elevation convention, unit, instrument or provider correction state, and uncertainty. A correction is applied only when the source record and selected operation establish that it has not already been applied. Unknown metadata are preserved as unknown and cause the corresponding derived quantity to be withheld; the code does not infer a datum or silently convert a likely unit. This fail-closed rule matters because a numerically plausible anomaly can be produced from an incorrect sign, a duplicated free-air term, a mixed reference datum, or a coordinate error.

For a station record i, the processed anomaly can be represented as the source observation plus a vector of signed corrections and a reference shift,

$$g_i^{\mathrm{proc}} = g_i^{\mathrm{src}} + \mathbf{c}_i^{\mathsf{T}}\mathbf{a} + \Delta g_{\mathrm{datum}}, \tag{3}$$

where $\mathbf{c}_i$ contains only corrections supported by the declared input and $\mathbf{a}$ contains their sign convention. This notation is an accounting identity, not an assertion that every source provides every correction. A missing correction value is not equivalent to zero. The implementation keeps the source and derived columns separate so a correction chain can be reproduced and audited. The no-double-correction validator rejects a record whose declared state is inconsistent with the requested chain before the output is emitted.

If source and correction errors are independent, the propagated variance is the sum of their contributions. More generally, the declared covariance is transformed by the same linear operation,

$$\sigma^2_{i,\mathrm{proc}} = \mathbf{a}_i^{\mathsf{T}}\,\mathbf{C}_i\,\mathbf{a}_i, \qquad \mathbf{C}_{\mathrm{proc}} = \mathbf{A}\,\mathbf{C}_{\mathrm{src}}\,\mathbf{A}^{\mathsf{T}}. \tag{4}$$

Here $\mathbf{a}_i$ is the local coefficient vector for the scalar record, whereas $\mathbf{A}$ is the complete linear map over a stacked survey. A diagonal-only uncertainty output is justified only when the input specifies independent errors or when the transform preserves that assumption. The processor does not manufacture cross-station covariance from marginal standard deviations.

Implementation is split between source parsing, unit and datum checks, correction-state validation, deterministic calculation, and a result receipt. The receipt binds each output to its parent asset and records the applied correction sequence, uncertainty provenance, and the implementation identity. The station tests exercise a known correction lineage, unit conversion, reference handling, uncertainty propagation, and rejection of duplicated or unknown operations. These tests prove the declared cases and code paths; they do not establish that an arbitrary downloaded table is a complete or eligible gravity survey.

The lesson is deliberately connected to the potential-field inverse that follows. A station table can pass syntax checks while remaining scientifically inadmissible because of an undocumented tide convention, unresolved datum, missing terrain model, or provider-level correction that cannot be reconstructed. The interface therefore separates raw records from processed records and exposes the provenance of each column. A user must be able to determine which values came from the provider, which were calculated locally, and which remain unavailable. No corrected anomaly is presented as subsurface density, and no correction step is described as an inversion. The relevant executable entry point is data-pipeline/gravity_processing.py; the stable station contract and rejection tests are in tests/numerics/test_gravity_processing.py.

The correction receipt also provides a practical way to compare processing paths without conflating them with a subsurface model. A reviewer can start from the same immutable station rows, select the permitted correction flags, and compare each derived column and uncertainty against a direct hand calculation. This check matters when adapting provider tables: field names alone do not reveal whether a Bouguer or terrain term is already included. The parser keeps the original numeric value and source metadata so the anomaly can be regenerated without overwriting the received record.

Coordinate conversion is handled as its own provenance step. It preserves the source coordinate reference and transforms station locations before any spatial derivative, gridding or prism sensitivity is evaluated. The transformed coordinates, selected local origin and axis order travel with the derived survey. A correction is never reconstructed from display coordinates rounded for a map. That rounding may be visually harmless and still unacceptable for a derivative or a high-gradient terrain model. The result contract therefore separates display formatting from the coordinates used in the forward calculation.

Weighted three-dimensional gravity prism inversion

The objective and depth weighting are grounded in established minimum-structure inversions for gravity and magnetics [21, 20]. The present mesh, data weights and calibration are product-specific and are not a reproduction of those published implementations.

The gravity inverse maps a discretized density contrast to observations through a forward operator assembled from three-dimensional prism responses. Let $\mathbf{m}$ contain cell density contrasts and let $\mathbf{G}$ map those contrasts to vertical gravity at the observation locations. For the declared linear density model, the prediction is

$$\mathbf{d}_{\text{pred}} = \mathbf{G}\mathbf{m}, \qquad \mathbf{r} = \mathbf{d}_{\text{obs}} - \mathbf{G}\mathbf{m}. \tag{5}$$

The units of $\mathbf{m}$, geometry, coordinates and $\mathbf{d}$ are checked at the interface. The mesh order is part of the numerical contract; a vector reshaped in a different cell order may look smooth but represents different physical cells.

The bounded L2 reference minimizes a noise-weighted data term together with a model-space penalty,

$$\Phi(\mathbf{m}) = \tfrac{1}{2}\|\mathbf{W}_d(\mathbf{G}\mathbf{m} - \mathbf{d}_{\text{obs}})\|_2^2 + \tfrac{\beta}{2}\|\mathbf{W}_m(\mathbf{m} - \mathbf{m}_{\text{ref}})\|_2^2, \qquad \mathbf{m}_{\min} \le \mathbf{m} \le \mathbf{m}_{\max}. \tag{6}$$

The reference model, bounds, weighting, regularization, mesh, active cells, and stopping conditions are reported as parameters of a run. Weights may encode stated standard deviations, but they do not transform a deterministic synthetic perturbation into a probability model. The minimizer is conditional on the mesh, reference and penalty. The reported model is not a unique geological image.

For a linear forward model the gradient and normal equations provide an independently checkable oracle,

$$\nabla\Phi(\mathbf{m}) = \mathbf{G}^{\mathsf{T}}\mathbf{W}_d^{\mathsf{T}}\mathbf{W}_d(\mathbf{G}\mathbf{m} - \mathbf{d}_{\text{obs}}) + \beta\mathbf{W}_m^{\mathsf{T}}\mathbf{W}_m(\mathbf{m} - \mathbf{m}_{\text{ref}}). \tag{7}$$

The project checks selected forward values against an independent prism calculation, then compares the SimPEG-backed solve with a bounded least-squares reference on the same data, survey, mesh and error model. This separates a forward-kernel error from optimizer configuration and from differences caused by changing the data or discretization.

Evaluation reports fitted and held-out predictions, weighted residual statistics, sensitivity or coverage diagnostics, and model-recovery metrics only when synthetic truth exists. The reference case demonstrates why these quantities are not interchangeable: a weighted WRMS near one coexists with a model correlation of 0.283 and a recovered centroid near 570 m for a truth near 282 m. The datum fit is consistent with the chosen scale, while the density distribution is unresolved. Potential-field depth/volume tradeoffs and mesh dependence remain; a single model slice cannot remove them. The implementation is data-pipeline/gravity_l2.py and the bounded calibration is covered by tests/numerics/test_gravity_l2.py and tests/numerics/test_gravity_l2_selection.py. The full family of solver outcomes, including ineligible or unresolved records, must be retained in the result catalogue rather than reduced to its best-looking case.

The choice of data weighting and regularization has measurable consequences. Multiplying every data standard deviation by a common factor changes the relative balance between fit and model penalty unless β is changed consistently. A comparison between two solvers is controlled only when observations, covariance, active cells, reference, bounds, regularization and stopping rules are shared. The locked L2 control matrix varies its declared case factors while keeping that comparison contract explicit. It records solver status separately from scientific recovery: termination proves that a numerical stopping rule was reached, whereas recovery requires an independent truth metric and can still fail.

The sensitivity matrix also explains why fit alone is insufficient for claiming resolution. Columns of **G** with similar predictions are difficult to distinguish in the measured geometry; the penalty selects one member from a family of acceptable models. A cellwise display should be read alongside data residuals, model regularization, depth sensitivity and synthetic truth when it exists. Potential-field depth/volume tradeoffs and mesh dependence remain. The browser may interpolate colors for display but must not regrid the model into apparent new resolution. This separation of mesh values from rendering coordinates is necessary for physical interpretation.

The reference model and active-cell mask are part of the mathematical problem, not cosmetic defaults. A different reference shifts the regularization center; a different mesh changes the columns and correlations of the sensitivity matrix; and a different standard deviation changes the relative contribution of each observation. Accordingly, the result record must preserve these values with the data identity and solver configuration. The independent prism calculation tests the forward sign and scale for its stated rectangular geometry, while the inverse comparison evaluates a separate question: whether the selected bounded, regularized model predicts the observed survey and recovers the known synthetic target. Neither calculation transfers to an arbitrary field survey without coordinate, datum, error, and source eligibility review. The L2 solution is therefore reported as a conditional estimate under a fully specified objective, not as the uniquely determined density distribution.

Aeromagnetic processing and harmonic continuation

The aeromagnetic workflow begins with flight-line geometry and provider leveling provenance. It checks coordinates, line identifiers, sample spacing, flight height, tie-line support, duplicate points, and data gaps before interpolation. A blocked flight line is reserved for evaluation before gridding; the withheld observations are not used to choose a grid or tune its smoothing. This test is intentionally different from randomly withholding nearby points, since spatially adjacent samples can share the same acquisition error and geological signal.

Potential fields in a source-free half-space satisfy Laplace's equation. For a horizontal Fourier component with wavenumber magnitude k, upward continuation by height h applies the transfer function

$$\widehat{T}(k_x, k_y, h) = \widehat{T}(k_x, k_y, 0)\,\exp(-h\sqrt{k_x^2 + k_y^2}). \tag{8}$$

Thus higher wavenumbers decay faster. The operation suppresses short wavelength content and is not an inverse continuation: recovering the surface field from an elevated grid would amplify noise. The implementation validates the transfer against a single analytic harmonic with known wavenumber and continuation height; the test is a physical identity, not an assertion that a gridded field uniquely determines its sources.

Interpolation and support are represented separately. Let $\mathcal{S}$ be the sampled footprint and $\mathcal{D}$ the output grid. A prediction outside an accepted support neighborhood is stored as unavailable, not silently filled with a value that looks measured. The gridding receipt records the interpolation rule, node spacing, coordinate frame, input line selections, mask, and any height transform. This permits a reviewer to distinguish the actual flight path from the raster display. The interface displays both the raster and the original lines, along with the blocked-line residuals and support mask.

The current public USGS Amargosa example uses the provider-levelled channel without repeating upstream diurnal, IGRF, navigation, or flight/tie-line levelling corrections.

Its rights and source hash are recorded in the research evidence; the calculation is a bounded processing/QC example, not independent site validation or a magnetic inversion. Upstream corrections cannot be reconstructed from absent instrument and processing records. A different provider dataset would need its own line geometry, units, leveling history, datum, and rights decision before being processed.

The source implementation separates parsing, line selection, blocked holdout, support-aware gridding, continuation, and output provenance. The core processing functions are in data-pipeline/magnetic_processing.py, while upward continuation and its independent transfer check are in data-pipeline/magnetic_continuation.py and tests/numerics/test_magnetic_continuation.py. The measurable acceptance evidence is the held-out line and analytic harmonic; a smooth map alone is not a processing validation. Flight-line texture, mask boundaries, physical units and the acquisition track should remain visible in scientific views so that interpolated values cannot be mistaken for direct observations.

Blocked-line scoring is calculated in the observation domain before interpretation of the raster. For held-out samples H, the noise-scaled residual summary is

$$\mathrm{WRMS}_H = \sqrt{\frac{1}{|H|}\sum_{i\in H}\left(\frac{T_i^{\mathrm{pred}} - T_i^{\mathrm{obs}}}{\sigma_i}\right)^2}, \tag{9}$$

provided each σ_i belongs to the source-backed or explicitly authored error model. If that uncertainty is unavailable, the report uses a raw residual distribution and labels it as such; it does not invent a noise scale. The blocked-line mask is checked against the fitting rows so a withheld line cannot enter interpolation selection through a separate code path.

The output should retain the flight path, elevation metadata, tie-line intersections and exact support mask. Height variation must not be flattened by silently assigning one nominal altitude. Continuation assumes a common upward shift in a source-free region; if receiver elevations differ, they must be accounted for by the declared geometry or a separate forward model. Finite-grid edge effects remain visible in the mask. An appealing color image that hides those boundaries can overstate the area constrained by measurements.

This path is narrower than a complete aeromagnetic interpretation. It does not recompute diurnal variation, IGRF removal, leveling, microleveling, reduction to the pole or source inversion. Those steps need the raw navigation, magnetic base-station, regional reference and survey metadata that define them. The product preserves upstream corrections when documented and refuses to label a provider-levelled series as raw. The view shows the input state and operations actually run, not a generic sequence of plausible-looking products.

Induced-magnetization prism modelling

The induced-field model assumes that each active cell's magnetization is parallel to a declared inducing field. For cell susceptibilities $\boldsymbol{\chi}$ and a fixed field direction $\hat{\mathbf{h}}_0$, the cell magnetization is proportional to $\chi_j \mathbf{H}_0$ under the stated linear, isotropic assumption. A prism kernel maps each cell to the vector magnetic field at receiver i; the scalar total-field response is a first-order projection onto the inducing direction,

$$\Delta T_i \simeq \hat{\mathbf{h}}_0^{\mathsf{T}} \Delta \mathbf{B}_i, \qquad \Delta \mathbf{B}_i = \sum_j \mathbf{K}_{ij}(\mathcal{G})\, \chi_j \mathbf{H}_0. \tag{10}$$

Here $\mathcal{G}$ includes prism bounds, coordinates, and the local frame. The approximation is the total-field linearization appropriate to the declared weak-anomaly model; the project

does not reinterpret it as a recovered vector magnetization. Remanent magnetization that is not parallel to the inducing field violates the assumption and is a negative-control case.

The fit objective is assembled from the physical forward predictions and a covariance model for the observation errors. For residual $\mathbf{r}(\mathbf{m}) = \mathbf{d}_{\text{pred}}(\mathbf{m}) - \mathbf{d}$,

$$\Phi(\mathbf{m}) = \tfrac{1}{2}\mathbf{r}^{\mathsf{T}}\mathbf{C}_d^{-1}\mathbf{r} + \beta\,\Phi_m(\mathbf{m}), \qquad \nabla\Phi = \mathbf{J}^{\mathsf{T}}\mathbf{C}_d^{-1}\mathbf{r} + \beta\nabla\Phi_m. \tag{11}$$

The covariance may contain correlated errors; replacing it by a diagonal weight matrix changes the likelihood and therefore the inverse problem. The executable assembly binds the survey, sensitivity, covariance, model parameterization, prior, source implementation and selected likelihood into one objective identity. Derivative tests compare the analytical gradient with finite differences and a separate Choclo field calculation on fixed geometries.

The currently retained 528-cell linear-TMI development fold has a source-bound objective and oracle certificate. It is a bounded synthetic fold, not completed survey inversion: candidate selection, sparse and nonlinear solves, held-out evaluation, field qualification, final export, and online resource admission remain unresolved. A converged fold would establish only that a specified numerical solve met its conditions; it would not validate remanence, geology, the inducing-field record, or performance on a real survey.

The front end and report must expose the receiver geometry, inducing field vector and coordinate convention, model units, active-cell mask, predicted total-field anomaly, residuals, objective contributions and solver status. The display must not show a single centered body as if it were the recovered geology. Independent prism checks and noise-covariance diagnostics have separate evidence from optimizer convergence. Implementation details are in data-pipeline/magnetic_forward.py, data-pipeline/magnetic_fit_assembly.py, and data-pipeline/magnetic_likelihood.py; source-bound tests include tests/data/test_magnetic_fit_assembly.py and tests/numerics/test_magnetic_forward.py.

Parameter scale is part of the objective identity. Susceptibility is dimensionless in SI, while the inducing field and resulting anomaly use field units; unit conversions applied to observations must not be applied a second time to the physical kernel. The residual covariance is constructed in the same component order as vectorized observations. A permutation of receiver/component indices changes both covariance and Jacobian correspondence, so the assembly validates their shapes and identity before solving. These structural checks do not replace independent kernel values; the Choclo comparison remains a distinct mathematical oracle.

Gradient evidence also has a limited scope. A finite-difference test checks a local derivative near selected model values and step sizes; it does not prove global convexity or an accurate optimum. A stationarity receipt must bind to the exact objective, covariance, source and model parameterization used. Candidate ranking and outer held-out evaluation require separate data. The passed development fold keeps its 72 validation and 72 outer rows sealed, preventing calibration or debugging from leaking those measurements into selection. The remaining work is not closed by extending iterations on one fold; acceptance requires source-complete selection, held-out scoring and reproducible export under the predeclared criteria.

For interpretation, the output distinguishes scalar total-field data from the vector secondary field used internally by the prism calculation. The inducing field amplitude, inclination, declination, coordinate frame, sign convention, model bounds and survey coverage must be visible. A sensitivity view can show which observations constrain a parameter, but it must not be labeled uncertainty unless a calibrated uncertainty procedure was run. This boundary prevents a forward calculation from being mistaken for a completed field inverse. The selected physical fold is also limited to one declared acquisition

and parameterization. Its nonzero response and derivative checks show that the assembled objective changes with the tested model and agrees locally with finite differences; they do not establish robust outer generalization, uncertainty calibration, or a field-ready magnetic inverse. The source, geometry, and inducing-field frame remain attached to every interpretation of its residuals.

Magnetotelluric tensor quality control

Phase-tensor theory supplies dimensionality diagnostics but does not make a necessary condition sufficient [22]. The product also records MTH5 as a time-series metadata and exchange reference [23]; it does not implement an MTH5 importer.

The MT tensor screen tests the observation object before a layered inversion is eligible. An EDI transfer function is already a processed frequency-domain product; the format and parser documentation is distinct from time-series storage formats such as MTH5 [14, 23]. It is not the original electric and magnetic time series. Its complex tensor, variance blocks, frequency vector, channel geometry, rotation declarations and optional tipper must therefore be internally coherent before a downstream earth model is interpreted. The parser in `data-pipeline/mt_field_qc.py` reads the original bytes, imposes a 10 MB bound, checks block counts and sentinels, and refuses ambiguous channel orientation, duplicated identifiers, incomplete remote-reference pairs and nonzero rotations that would require transforming unavailable covariance. The independent `mt_metadata` reader must agree with the hand-read frequency, impedance and squared component-error arrays. This is a parser comparison, not evidence that the upstream time-series processing was optimal.

For a uniform isotropic one-dimensional earth, the diagonal tensor terms vanish and the off-diagonal terms are antisymmetric in a correctly aligned frame. The screen measures departures from these necessary conditions against the source-provided component variances. For each frequency k, it calculates normalized diagonal and antisymmetry scores,

$$q_{xx,k} = \frac{|Z_{xx,k}|}{\sqrt{2}\,\sigma_{xx,k}}, \quad q_{yy,k} = \frac{|Z_{yy,k}|}{\sqrt{2}\,\sigma_{yy,k}}, \quad q_{a,k} = \frac{|Z_{xy,k} + Z_{yx,k}|}{\sqrt{2}(\sigma_{xy,k} + \sigma_{yx,k})}, \qquad Q_c = \sqrt{\frac{1}{N_f}\sum_{k=1}^{N_f} q_{c,k}^2}. \tag{12}$$

The configured necessary-screen limit is 3.0 for each component family. For the antisymmetric reference tensor J, the relative distance

$$d_k = \frac{\|Z_k - a_k J\|_F}{\|Z_k\|_F}, \qquad a_k = \frac{Z_{xy,k} - Z_{yx,k}}{2}, \tag{13}$$

is also recorded. A quarter-turn of both axes must leave this distance unchanged to absolute error 10^{-12}. This control detects an implementation or frame error; it does not license arbitrary rotation of the measured tensor.

The phase-tensor diagnostic uses $\Phi_k = \Re(Z_k)^{-1}\Im(Z_k)$ only when the real part has finite condition number no larger than 10^8. It reports the median and 90th-percentile scalar departure, skew angle and skipped periods; fewer than half numerically usable periods fails the diagnostic. A tipper is checked as a paired complex vector with a common missing-value mask, and its median magnitude is descriptive. The calculation makes no geological classification from these summaries. Covariance between tensor entries is not inferred, missing periods are not silently imputed, and unresolved units or sign conventions keep an absolute inverse ineligible.

The executed field intake is AusLAMP NSW C15, whose EDI and provider screening records are separately hash-bound. Its measured transfer function is not presented as raw field time series, and the source EDI does not declare all physical conventions

needed for an absolute inverse. The product therefore records QC while keeping inversion disabled. The separate Clear Lake station cl061 also remains QC-only because its necessary isotropic-1D consistency screen fails. A passing necessary screen would still not prove that the subsurface is one-dimensional, that static distortion is absent, or that phase-tensor dimensionality is unique. This is an eligibility and audit stage, not a solver or a truth estimator. The online contract preserves this boundary by requiring an MT-screen results to have no inverse object and to state that inversion was not performed. This fail-closed result allows the interface to show measured curves, missing masks and reasons without inventing a layered model.

The score is deliberately described as a necessary tensor null rather than a station classifier. The generous variance treatment divides by the square root of each native variance for diagonal components and combines marginal standard deviations for the antisymmetric pair. Since EDI variance conventions and cross-component covariance are not always specified, this bound can admit data that a stricter covariance-aware test would reject. The receipt records the convention verbatim and marks the physical interpretation unresolved. The screen uses every supplied period and does not choose a favorable band after viewing the score. The quarter-turn invariance check is applied to the relative tensor distance and is not a substitute for a full coordinate rotation with uncertainty propagation. This restriction is why nonzero source rotation is refused for this candidate instead of normalized by silently rotating the tensor.

The displayed apparent-resistivity and phase curves are useful for human inspection, but neither curve replaces the complex tensor in QC. Phase wrap, low-amplitude components and sign convention can make visually similar curves correspond to different complex data. Missing tipper periods are shown as gaps, not interpolated points. The provider's reported one-dimensional class is attributed to its pinned dimensionality record, while the independently computed scalar departure is shown separately; disagreement is evidence to investigate, not a license to override either record. In the app, selecting a station or component must update the frequency plot, tensor diagnostics and reason list from the same source-bound receipt. This design keeps observation-level QC distinct from any later inverse result.

Layered-earth magnetotelluric forward and inverse

The layered-earth inversion uses a quasi-static plane-wave response for horizontally layered, isotropic earth. A model contains n positive resistivities and $n-1$ finite layer thicknesses; the bottom half-space has no finite thickness. With the declared $\exp(+i\omega t)$ convention and $\mu_0 = 4\pi\,10^{-7}$ H/m, the basal impedance and upward recursion are

$$Z_n(\omega) = \sqrt{i\omega\mu_0\rho_n}, \tag{14}$$

$$k_j = \sqrt{\frac{i\omega\mu_0}{\rho_j}}, \quad w_j = \sqrt{i\omega\mu_0\rho_j}, \quad t_j = \tanh(k_j h_j), \tag{15}$$

$$Z_j(\omega) = w_j \frac{Z_{j+1}(\omega) + w_j t_j}{w_j + Z_{j+1}(\omega) t_j}, \qquad j = n-1, \ldots, 1. \tag{16}$$

The forward model is evaluated at each measured frequency. Apparent resistivity and phase are derived observables,

$$\rho_a(f) = \frac{|Z(f)|^2}{\mu_0 2\pi f}, \qquad \varphi(f) = \operatorname{atan2}(\Im Z(f), \Re Z(f)). \tag{17}$$

The interface must label phase in degrees and resistivity in ohm metres; the complex impedance is primary to fitting, while the two curves are readable transformations of

that complex value.

The product compares bounded Levenberg-Marquardt, Adam-based and neural baselines on controlled cases. Resistivity is parameterized by its logarithm, fixed thickness is supplied as input, and every layer is bounded strictly inside 1 to 6000 ohm metres. With active-frequency mask m_k, observed impedance d_k, per-real-component uncertainty σ_k, and roughness weight β, the reported objective is

$$J(x) = \frac{1}{2N_a} \sum_{k:m_k=1} \frac{|Z_k(\exp x, h) - d_k|^2}{\sigma_k^2} + \frac{\beta}{n-1} \sum_{j=1}^{n-1} (x_{j+1} - x_j)^2, \tag{18}$$

with the second term defined as zero for a one-layer model. The real and imaginary residuals use the same component standard deviation; this is not a covariance model for correlated tensor components. Inactive frequencies are excluded from the objective and retained in the output as held-out predictions. Parameter changes recompute the complex response and its curves, rather than shifting a pre-rendered plot.

Local identifiability is assessed from finite-difference sensitivities with respect to log resistivity. Singular values, effective rank, weak parameter participation, a local linearized covariance and proximity to bounds are exported. These are conditional local diagnostics with layer thickness fixed; they are neither a posterior distribution nor a proof of uniqueness. A seeded parametric bootstrap uses the declared Gaussian real/imaginary noise law and reports empirical coverage with Monte Carlo standard error and Wilson intervals. Those intervals quantify this specified model and noise law only. They do not include uncertainty in thickness, dimensionality, static shift, source processing or geological structure.

The API validates that a field inverse is tied to an immutable QC result, fixed thickness and the exact source hash. It replays the screen before accepting the layered-earth job and checks the resulting environment and solver receipt. Clear Lake cl061 is never promoted from QC to inversion because its MT tensor screen fails. The C15 record is also QC-only while units, variance convention and time-sign convention remain unresolved. Thus the online calculator is demonstrated for eligible synthetic or fully qualified submissions; a successful numerical fit alone cannot make an ineligible EDI scientifically admissible.

Log parameters enforce positive resistivity and make proportional changes in each layer more comparable during optimization. Fixed thickness is an important experimental constraint: this method does not jointly estimate layer boundaries in the reported exercise. The regularizer penalizes adjacent log-resistivity differences, so increasing β favors a smoother layered sequence and can suppress thin conductive or resistive features. The chosen β is therefore reported with the starting model, bounds, active frequencies and parameter count. If $\beta = 0$, weakly resolved layers may vary along near-null sensitivity directions; a small data residual does not resolve that ambiguity.

The forward recursion is linear in the number of layers for each frequency, while each nonlinear objective evaluation scales with both the layer count and number of active frequencies. Numerical derivatives involve repeated forward calls, and the local uncertainty calculation requires a singular-value decomposition of a Jacobian with one column per layer. These bounded one-dimensional problems are inexpensive enough for local replay, but the code still caps layer dimensions and bootstrap work in the API contract. Every result records the active mask and the held-out response separately; held-out samples cannot contribute to the regularization or parameter update.

The comparison notebook shows the complex data fit, derived apparent resistivity and phase, model profile, per-frequency residual and identifiability diagnostics on linked axes. A user can change layer resistivity and fixed thickness, rerun the physical recursion and

inspect how a deeper layer affects lower frequencies. The same control changes the prediction and residual, making the mapping from model to response auditable. It does not expose a free-text model as if it were a validated field interpretation. If an uploaded EDI fails the MT tensor screen, the interface retains the QC result and disables the layered-earth inverse; it does not substitute the nearby synthetic example or fabricate an inverse.

Electrical resistivity tomography with topography

The modeling and inversion context includes the open pyGIMLi library [27]. The local user-data adapter and its mesh/topography limits are described separately below.

The ERT workflow processes a provider-linked two-dimensional Wenner profile in the local isolated pyGIMLi environment. Its source-specific `.ohm` reader accepts at most one megabyte, requires the declared provenance and Wenner header, metre-valued electrode coordinates, resistance in ohms, strictly increasing surface sensors and exact ABMN indices. It rejects duplicate quadrupoles, malformed tails, nonfinite values and unsupported geometry rather than inferring columns or units. The immutable source hash and provider-link-only rights status remain attached to the local receipt; field truth is explicitly null.

For a flat homogeneous half-space, the signed four-electrode factor gives an independent check on the library geometry. With electrode positions x_A, x_B, x_M, x_N,

$$K_{ABMN} = \frac{2\pi}{|x_A - x_M|^{-1} - |x_A - x_N|^{-1} - |x_B - x_M|^{-1} + |x_B - x_N|^{-1}}, \qquad \rho_a = K_{ABMN} R_{ABMN}. \tag{19}$$

The analytical factor is compared with pyGIMLi's flat-earth factor to relative error at most 10^{-5}. The topographic numerical factor is then computed from the supplied surface geometry; it must be finite, positive and shape-consistent. This separates the analytic geometry oracle from the topographic correction the inverse actually uses.

QC preserves every observation and does not silently remove outliers. Reciprocal pairs are identified by reversed ABMN indices. Within each electrode-spacing class, rows whose absolute log-resistance deviation exceeds six scaled median absolute deviations are flagged, but the flags do not alter weights. Instrument errors are marked not supplied. The primary split selects every fifth measurement after sorting by spatial midpoint within AB-index span; a second, spatially blocked split removes the central 40–60 percent segment. Both rules are independent of the observed resistance. The software checks that pyGIMLi preserved the strictly parsed coordinates, indices and resistance values before either split can be inverted.

The conditional inverse builds a topographic finite-element mesh and uses log-resistance predictive error on rows held out from fitting. A median apparent-resistivity baseline is evaluated on the identical held-out rows. The solver uses $\lambda = 10$, parameter spacing 0.3, maximum cell size 10 m, depth 20 m, mesh quality 33.6 and at most 12 iterations. Sensitivity reruns perturb error scale and mesh resolution without selecting the primary model after seeing held-out performance. The log-resistivity RMSE is

$$E_{\log R} = \sqrt{\frac{1}{N_h} \sum_{i \in H} [\log(\widehat{R}_i) - \log(R_i)]^2}, \qquad \Delta = 1 - \frac{E_{\log R,\text{inverse}}}{E_{\log R,\text{baseline}}}. \tag{20}$$

The acceptance rule requires an ordinary engine stop and at least 10 percent held-out improvement over the declared baseline; reaching the iteration limit is not convergence.

Mesh hashes use rounded physical coordinates and cell markers, not node identifiers, so a rerun can distinguish a true geometric change from library numbering.

This pipeline is local and source-specific, not a general ingestion API for arbitrary ERT formats. A restricted or unavailable field file is not fabricated as an executed example; the committed synthetic control is used to check the numerical route separately. Positive resistances are required for the log objective, and the one-dimensional Wenner survey cannot establish three-dimensional geology. Mesh dependence, topographic factor sensitivity, reciprocity and spatial holdout are therefore reported beside the recovered section. A smooth section is a regularized model conditioned on geometry, bounds and error assumptions, not a uniquely resolved image of the subsurface.

The field profile itself has a single line and a provider-declared Wenner acquisition. It therefore supports a conditional 2.5D sensitivity exercise, not arbitrary three-dimensional electrode geometry. The geometry factor convention is checked before inversion because an incorrect sign or unit would propagate into every apparent-resistivity value. The log-space comparison is used because a fixed additive resistance difference has different physical weight across orders of magnitude. The held-out RMSE is computed in the same native resistance space after log transformation; it is not a posterior likelihood.

The mesh parameters are frozen in the primary run so analysts cannot select the displayed model by repeatedly changing the mesh after looking at the held-out segment. Alternate mesh and error-scale runs are sensitivity records, not extra independent validation cases. The blocked segment is a spatial stress test because adjacent Wenner quadrupoles are strongly related; the every-fifth split alone can leave nearby geometry in both partitions. Agreement across both splits is more informative than a random row split, but neither split makes the field survey representative of unobserved lithology.

The local receipt records sensor coordinates, ABMN rows, apparent resistivity, topographic and flat factors, reciprocal pairs, QC flags, training and held-out identities, mesh hash, option values, convergence reason and prediction arrays. Plotting must preserve topographic elevation and actual electrode spacing. A color-filled pseudosection is an apparent-data view, not the inverted earth model; the model section must be rendered with its mesh and parameter units. If the solver fails to stop normally or the held-out improvement is below the predeclared threshold, the model remains visible as a failed or unresolved calculation and is not labeled accepted.

Waveform response, filtering and classical picks

Waveform processing begins with exact MiniSEED bytes and a StationXML inventory. The request identifies network, station, location, channel, UTC interval and processing parameters. The parser checks record boundaries, sample-rate and timing continuity, NSLC identity, declared time coverage, channel orientation and response epoch. Inventory response is resolved over the entire requested interval; a gap, epoch mismatch or unsupported response stage fails closed. This prevents a channel's sensitivity at a nearby time from being borrowed to justify response removal at the sample time. The raw files, request identity, software versions, response-work units, filter coefficients, output descriptors and QC reasons are hash-bound in the local result. External helper calculations do not claim filesystem custody unless a separate execution receipt proves it.

After demeaning and validated response removal, a zero-phase Butterworth bandpass is applied with second-order sections and forward-backward filtering. The order, corner frequencies, padding length and coefficients are part of the result. Welch spectra use a periodic Hann window, 50 percent overlap, constant detrending and density scaling. For a window of length m at sample rate f_s, the spectral density and normalization are

recorded with segment count, frequency increment, omitted tail and integrated RMS. The result labels spectral summaries as diagnostics rather than calibrated noise or magnitude.

The classical picker computes the short-term average to long-term average ratio (STA/LTA) on the processed trace. Before this ratio is formed, samples are scaled by the largest absolute amplitude to avoid overflow without changing the ratio. For window lengths $n_s < n_l$,

$$C_i = \frac{n_l^{-1} \sum_{j=i-n_l+1}^{i} x_j^2}{n_s^{-1} \sum_{j=i-n_s+1}^{i} x_j^2 + \epsilon}, \qquad \text{trigger on } C_i > \theta_{\text{on}}, \quad \text{close on } C_i \le \theta_{\text{off}}. \tag{21}$$

The executable uses ObsPy's classic STA/LTA characteristic and a hysteresis state machine: each interval records its first active sample, maximum-ratio sample, exclusive stop, channel, UTC time, refractory interval and whether valid data truncated either edge. Empty or zero traces yield no fabricated trigger; candidate counts are bounded.

Where analyst picks exist, evaluation compares matched channels and windows without letting the catalog reference change preprocessing or selection. A pick is judged within a declared time tolerance, with misses, false picks, outside-tolerance detections and unlabelled candidates counted separately. The method does not identify P or S from energy ratio alone; phase remains null unless a reviewed phase rule supplies it. Instrument correction depends on accurate StationXML and epoch matching, and a bandpass can shift apparent onset or suppress signal. STA/LTA is therefore a reproducible candidate generator for inspection, not an automatic earthquake catalog, calibrated arrival uncertainty, or field accuracy claim. User traces remain client-side unless the owner submits them to the protected waveform job service under the separate account and resource contract.

The fixed candidate threshold and STA/LTA windows are stored in the submitted request, rather than guessed from the most prominent visual peak. For sample interval Δt, the sample lengths are obtained by rounding $0.12/\Delta t$ and $1.2/\Delta t$, with at least two short samples and one additional long sample. The implementation rejects windows shorter than three long-term windows. The onset detector used in the STEAD baseline requires an actual upward threshold crossing; a maximum selected within a known analyst window would leak the reference arrival into the prediction. The product keeps this baseline distinct from the response-corrected measured waveform path.

For a trace x_n, the one-sided Welch power spectral density is computed by averaging windowed segment periodograms,

$$\widehat{S}_{xx}(f) = \frac{1}{K f_s \sum_{n=0}^{m-1} w_n^2} \sum_{k=1}^{K} \left| \sum_{n=0}^{m-1} w_n x_{n+kq} e^{-2\pi i f n / f_s} \right|^2, \qquad q = m/2. \tag{22}$$

The exact implementation uses a periodic Hann window and records the segment count, tail omission and integral used for the RMS diagnostic. This equation describes the spectral estimator; it is not a noise probability model. The high-pass and low-pass edges must remain below Nyquist, and records too short for stable padding are rejected rather than filtered with a different implicit rule.

The waveform result has three distinct layers: parser/QC status, processed physical arrays and candidate picks. A channel can be readable while response removal remains unavailable, and a computed candidate list can exist while field eligibility remains false. The browser should show original and processed traces with matching UTC axes, mark gaps and response epochs, then overlay threshold intervals. The same selected channel drives its PSD and pick list. These linked views allow inspection of filter effects and candidate timing without conflating an attractive trace with a correct phase classification. Observed picks and independent truth remain separate records. When a reference catalog is unavailable, accuracy is not scored; the result reports descriptive diagnostics and an explicit unresolved status.

First-arrival travel-time tomography

First-arrival functional and eikonal formulations, including irregular-surface treatment and crustal tomography, are established methods [24, 30, 31, 32]. The local implementation does not claim general 3D or topography-aware equivalence.

First-arrival tomography estimates a two-dimensional velocity field from picked arrivals and times and explicit source-receiver coordinates. The local source adapter reads a bounded pyGIMLi `.sgt` survey and preserves the original bytes, their SHA-256, provider or user-declared provenance, coordinate frame, units, pick values and rights decision. Source positions and receivers must be finite, distinct and within the declared model domain. Repeated paths are not silently combined. User-supplied local surveys must explicitly declare metre coordinates, seconds, and the vertical axis convention. A y-down input is transformed to the engine's y-up frame, and both source and transformed-input hashes are retained. Without a declared frame or unit basis, the inverse is rejected.

The straight-ray discretization represents each observation with path lengths G_{ij} through cells j. If $s_j = 1/v_j$ is slowness, the predicted time and residual for ray i are

$$t_i^{\text{pred}} = \sum_{j=1}^{N_c} G_{ij} s_j, \qquad r_i = t_i^{\text{obs}} - t_i^{\text{pred}}. \tag{23}$$

This matrix view makes the linearized ambiguity visible: rays that do not cross a cell provide no direct path-length sensitivity there. Ray coverage is exported alongside the velocity, and uncovered or weakly covered cells are not described as resolved structure. The product's field operator uses pyGIMLi travel-time simulation with a declared secondary-node discretization; the independent linearized weighted least-squares implementation is a comparison, not a replacement oracle for bent-ray physics.

Training and evaluation are grouped by complete shot, never by individual ray from a shot. The frozen policies include every fifth shot in survey order and a central-shot block. Picks from every held shot are excluded from parameter fitting, and a fresh forward manager rebuilds the acquisition graph on the fitted mesh before predicting them. A training-only homogeneous slowness is obtained from the weighted least-squares slope through offset and time,

$$\widehat{s}_0 = \frac{\sum_{i \in T} w_i x_i t_i}{\sum_{i \in T} w_i x_i^2}, \qquad w_i = \sigma_i^{-2}, \qquad \widehat{v}_0 = \widehat{s}_0^{-1}. \tag{24}$$

The baseline and mesh inverse are scored on the same held-shot rows. The current source defaults the per-pick scale to the greater of three percent of the observed time and 0.1 ms. The inner CGLS solve is pinned to 1000 iterations and absolute squared-residual tolerance 10^{-20}; the outer mesh, starting model, z weighting and iteration budget are recorded in the receipt. The independent forward replay must match the fitted training response before held-shot error is accepted.

The key limitation is the straight-ray geometry used by user-input velocity calculation and several comparison paths. It neglects ray bending, anisotropy, finite-frequency sensitivity, uncertain picks and three-dimensional structure. The source-specific pyGIMLi path and user-provided local-grid path are separate contracts; neither assigns a CRS or vertical datum from coordinate magnitudes. A lower travel-time residual does not establish recovered geology. Method status depends on held-shot improvement, solver stop state, coverage, mesh sensitivity and independent replay, not solely on the training objective. Result exports include observed and predicted times, residuals, cell centers, mesh identity, coverage, split identities, engine versions and the explicit no-truth field boundary.

For source data from pyGIMLi, the file does not declare every physical convention. The adapter records the documented interpretation of its local coordinates and seconds rather

than treating that interpretation as provider-authored metadata. The user-data path is stricter: provenance must identify the owner-declared source and licensing basis, and the vertical direction is explicit. A coordinate sign conversion changes the engine input hash but leaves the original source hash intact. The report therefore distinguishes the bytes analyzed by the solver from the exact bytes supplied by the user, and reports that CRS and vertical datum were not independently verified.

Mesh cells with low ray-path length coverage can acquire extreme values under smoothness regularization even when their effect on fitted data is negligible. The coverage array and parameter mesh are thus part of the result contract, not optional plot metadata. A map that paints every cell with equal opacity hides this distinction; the display should overlay coverage or visibly mask unsupported cells. Residuals are shown against offset and by held shot, because a pooled RMSE can conceal one poor acquisition geometry. The direct-path model is most defensible when velocity contrasts and offsets support approximately straight first-arrival rays. Strong refraction, shadow zones or anisotropic propagation require a more appropriate forward model before a velocity cross-section is interpreted.

The frozen split manifest contains ordered shot identifiers and hashes the exact training and held-out row lists. On replay, the parser verifies that pyGIMLi has not reordered those rows or reintroduced a held shot into the fitting container. The independent forward operator then predicts the complete acquisition on the fitted mesh and checks the training portion against the inversion's stored response. These identity and operator checks make a low held-out error attributable to the declared model and partition rather than a split leak or stale response. Uncertainty remains assumed from the supplied scale; the receipt does not call it calibrated if no independent calibration experiment was run.

Acoustic full-waveform inversion

The multiscale rationale is established in the FWI literature [25]; the product tests two fixed schedules on bounded acoustic synthetic cases.

Acoustic FWI solves a two-dimensional, constant-density wave equation with a known source wavelet and explicit shot-receiver-time geometry. The reference experiment uses a 128×96 velocity grid, three source locations, recorded shot gathers and a controlled synthetic geological model. The forward operator runs through Deepwave, with discretization refinement and an adjoint-gradient control compared with an independently constructed reference. Observations have axes $[\text{shot}, \text{receiver}, \text{time}]$; a receiver holdout is applied consistently to every shot so the fitting operator and generalization measure are not confused.

For density ρ, velocity v, pressure p, source q and time t, the acoustic pressure equation can be written

$$\frac{1}{K(\mathbf{x})}\frac{\partial^2 p}{\partial t^2} - \nabla \cdot \left(\frac{1}{\rho(\mathbf{x})}\nabla p\right) = q, \qquad K(\mathbf{x}) = \rho(\mathbf{x})v(\mathbf{x})^2. \tag{25}$$

Here density is held constant in the product experiments, so the inversion parameter is velocity and not an independently estimated elastic property. With observed data d, synthetic response $F(v)$, fitted receiver set A, a low-pass operator L_f, and spatial roughness $R(v)$, a continuation stage minimizes

$$J_f(v) = \frac{\|A[L_f F(v) - L_f d]\|_2^2}{\max(\|AL_f d\|_2^2, \epsilon)} + \beta R(v). \tag{26}$$

The discrete gradient is obtained by backpropagating through the wave-propagation

graph. A directional finite-difference check and a refined-grid adjoint test are distinct controls: one evaluates the implemented derivative locally, the other checks the propagation discretization against a separate implementation.

The matched comparison first estimates a shared one-dimensional low-frequency background from training observations, then gives the ordinary L2 and multiscale schedules equal data, model parameterization, regularization and optimizer-call budgets. The multiscale schedule uses 3, 5, 8 and 14 Hz low-pass stages; the ordinary schedule uses the same spatial stages but omits the intermediate frequency continuation. Control grids are refined through four declared resolutions. L-BFGS uses a strong-Wolfe line search with one accepted update per call, history size 15, and explicit gradient and change tolerances. Every post-update state, including the final one, is recorded; saved image frames carry the exact history index so the interface does not animate a stale state as the terminal result. The held receiver set is never used to select a schedule, stopping time or checkpoint.

Cycle skipping occurs when the predicted phase is too far from the observed phase for local waveform correlation to point toward the correct model. Low-frequency continuation and a data-derived initial background mitigate this risk but do not remove it. The product also reports initial, active and held-out relative errors; residual spectra and receiver-resolved energy are diagnostics, not method rankings. Synthetic truth permits model-recovery metrics for the authored case, while user-provided data carry no true velocity field and receive no truth-based score. Constant density, acoustic physics, 2D geometry, source signature and finite-band data bound the interpretation. The experiment therefore teaches FWI's dependence on initialization, frequency and acquisition, not a field-scale claim or a general seismic migration result.

The finite-difference time step and spatial grid must satisfy the stability bound of the discrete acoustic scheme; the forward module records spacing, sampling interval, source positions, source spectrum and solver versions. The independent-start model is shared between competing schedules, while the one-dimensional background is estimated from training receivers only. This controls the most consequential initialization difference without claiming that a 1D background is adequate for a laterally varying target. The velocity parameterization uses a bounded control grid and interpolates it to the propagation grid; coarse controls reduce the number of independent parameters early in the optimization and are refined in later stages.

The fitted receiver mask omits every fifth receiver with a fixed remainder. Its complement is used only for final predictive diagnostics. At every saved iteration, the artifact retains the full model, its predicted shot gather, active and withheld relative errors, objective components and control-grid identity. Each plotted frame is linked to the recorded iteration index. This permits a reader to inspect how the wavefront and the inferred velocity evolve during the actual optimizer sequence rather than viewing a decorative animation. If only sparse frames are retained, the interface must label the actual frame indices and not interpolate solver states that were never computed.

The main optimization comparison does not use synthetic truth to choose its selected iterate. Truth is available only after fitting to compute model-recovery metrics and to diagnose the controlled case. This distinction prevents the best-looking model from being selected with knowledge of the answer. The objective normalization uses the active data amplitude so relative errors can be compared across cases, but the per-case amplitude and absolute residuals remain necessary for physical context. A low normalized objective on a weak trace does not imply a small absolute timing or amplitude error.

The manuscript reports a solver call as an optimization update only when the saved post-update model and prediction match the same terminal state. Adjoint correctness is tested against finite differences on declared directions and against a refined-discretization ref-

erence; these checks validate derivatives and discretization for the tested fixtures, not every wavefield, grid or boundary condition. The current method remains a synthetic acoustic comparison. User-supplied seismic traces require their own response, geometry and source-wavelet contract before this controlled workflow can be called a field inversion.

Joint potential-field inversion and structural controls

Structural joint inversion has a prior literature [26, 10, 11]. The report keeps structural coupling distinct from its separate mixture-prior branch.

Joint potential-field inversion estimates density contrast and magnetic susceptibility on a shared three-dimensional mesh. Its central scientific question is whether spatial coupling adds predictive value when the two modalities do not share the same geological boundary. The comparison retains three separate models: independent single-physics fits used as initial conditions, an optimized uncoupled joint reference with both data objectives, and coupled alternatives using a cross-gradient or a Gaussian-mixture petrophysical term. This arrangement makes the uncoupled two-modality fit the comparator; comparing a joint estimate only with one modality would confound added data with added coupling.

For property models m_g and m_m, the cross-gradient is

$$\mathbf{c}(\mathbf{x}) = \nabla m_g(\mathbf{x}) \times \nabla m_m(\mathbf{x}), \qquad J_c = \frac{1}{|\Omega|} \int_\Omega \frac{\|\mathbf{c}(\mathbf{x})\|_2^2}{s_c}\, d\mathbf{x}. \tag{27}$$

The vector vanishes when local gradients are parallel, including when one gradient is zero; it does not require equal property amplitudes. That makes it useful for structural alignment but insufficient to identify either property scale or sign. The implementation uses physical mesh spacing in the discrete Torch gradient, normalizes by a scale derived from the matched initial cross-gradient, and exports the component images for inspection. Cross-gradient weight is varied only in the predeclared calibration design.

The petrophysical alternative fits a two-component Gaussian mixture to 640 independently generated synthetic paired density and susceptibility samples, using seed 68121 and 80 expectation-maximization updates. The samples are not labels from the displayed volume, measured cores or field rock properties. In normalized property coordinates u, the penalty is the negative log mixture density

$$J_p = -\frac{1}{N} \sum_{i=1}^{N} \log \left[\sum_{k=1}^{K} \pi_k \mathcal{N}(u_i \mid \mu_k, \Sigma_k) \right]. \tag{28}$$

Both couplings are added to the same whitened gravity and magnetic misfit and spatial regularization terms. The separate term ledger records data, smoothness, cross-gradient and mixture contributions so that one favorable aggregate objective cannot hide a poor modality fit.

Optimization uses diagonal-Hessian preconditioning with L-BFGS and strong-Wolfe line search; the default comparison records 80 outer update calls. Density and susceptibility are scaled for numerical conditioning and unscaled for output. The covariance model, active-cell geometry, noise scales, mesh, inducing field and source coordinates are fixed by the case artifact. Fitted and withheld observations are separated by the stored mask. The evaluation reports each modality's data residual, held-out residual where available, model error against the synthetic truth, the coupled-to-uncoupled ratios and the reason-coded verdict. Known conflicting-boundary controls deliberately make structural coupling inappropriate; those outcomes are labeled negative controls, not optimizer failures to conceal.

A common structure is an assumption, not a universal geophysical law. Cross-gradient cancellation can occur in uninformative regions, and a learned mixture prior can pull both models toward a false shared population when the geology conflicts. The product therefore reports uncoupled and coupled outputs side by side and preserves the structural conflict result. The implementation is a research comparison on authored synthetic cases. The report distinguishes the independent potential-field operators from the joint objective and optimizer, and does not imply that the coupled update itself is a stock SimPEG cross-gradient inversion until the exact production operator is verified against that package. No field joint inverse or petrophysical calibration is claimed.

The gravity and magnetic sensitivity matrices are assembled from SimPEG's three-dimensional integral simulations using the GeoAna engine; the joint parameter updates are then performed by the product's preconditioned Torch objective. This distinction matters for reproduction: citing the forward engine does not make the coupling optimizer identical to SimPEG's built-in cross-gradient implementation. The mesh and spacing are shared so the finite-difference gradients act on corresponding physical cells. Active observations, cell weights, per-modality uncertainty, inducing-field direction and objective normalization are fixed in the case receipt.

The uncoupled run is optimized jointly with both independent data terms, so it is a stricter comparator than simply concatenating two separately inverted volumes. The coupled runs start from that same independent initialization and alter only the coupling term and its declared weight. Saved objective histories separate gravity fit, magnetic fit, spatial regularization, cross-gradient energy and mixture likelihood. Every terminal result is re-evaluated with the same simulation matrices and independent synthetic truth; parameter recovery and predicted-data agreement are reported separately because one can improve while the other degrades.

The synthetic prior is fitted independently of the displayed case, which avoids directly using the answer volume as a prior training set. It remains a generated two-population assumption rather than a learned rock-physics law. The deliberately conflicting geometry is a necessary counterexample: a coupling penalty can make the models more structurally similar while worsening one or both physical properties. In such a case, the appropriate recommendation is to inspect the uncoupled models and retain a negative-control status, not to increase coupling until the sections look aligned.

Learned first-arrival velocity estimation

Learned inversion and continuous neural parameterizations are established comparison families [28, 29, 12]; the local residual model is not a reproduction of those methods.

The learned-velocity study evaluates a learned refinement on controlled straight-ray first-arrival data. The input contract is a set of local x, z coordinates in metres, positive arrival times in seconds, and optional per-ray uncertainty. Each row becomes a source-receiver path; duplicate undirected paths, zero-length rays, nonfinite values and coordinates outside the fixed 0–800 m model domain are rejected. The physical forward comparison uses path lengths through a 16×16 grid and an independent bilinear quadrature operator. This explicitly separates the grid-aligned training operator from a second numerical integration used for evaluating predicted times.

For reference velocity v_0, path matrix A, ray length $\ell_i = \sum_j A_{ij}$, cell coverage $c_j = \sum_i A_{ij}$, and observed times t_i, the physics-conditioned features are

$$r_i = \frac{t_i - (A v_0^{-1})_i}{\ell_i}, \qquad h_j = \frac{\sum_i A_{ij} r_i}{c_j} \ (c_j > 0), \qquad f_j = -\frac{h_j v_{0,j}^2}{400}. \tag{29}$$

The second channel is normalized coverage $c_j / \max_k c_k$; uncovered cells receive a zero residual feature and remain identifiable in the coverage image. The CNN returns a bounded velocity model, while a separately selected regularized weighted-L2 solve remains the classical comparator. Training uses disjoint procedural families and a validation set for checkpoint selection; test cohorts are separated by in-distribution, family shift, acquisition shift and their joint shift. The training notebook and receipt bind model bytes, data split, random seed, optimizer history and evaluation files.

The frozen experiment uses 800 training and 160 validation examples, with test groups of 160, 80, 80 and 160 examples respectively. The current physics-v2 study trained for 40 epochs and selected the recorded validation checkpoint. On the 160-case joint family/acquisition holdout, the primary run's learned-to-classical velocity RMSE ratio is 2.194 and the independent travel-time RMSE ratio is 5.585. A same-seed CUDA replay gives ratios 2.190 and 5.595. All 160 cases exceed each run's validation-derived diagnostic threshold. The repeated seed measures execution sensitivity, not independent-seed variance. These are negative results for the frozen synthetic generator and operators.

The user-data function accepts at most 4096 unique paths and checks coordinates, time units, bounds, path identity and checkpoint hash before importing the heavier numerical and neural engines. Classical inversion and learned inference are both returned with observed, predicted, residual and normalized-residual arrays plus an independent quadrature prediction. Inference is explicitly exploratory and marks training geometry/noise-domain mismatch; it does not auto-calibrate the model for new acquisition. Straight rays, fixed reference structure, bounded velocity and absence of field labels limit the result. The experiment supports a reproducible negative comparison, not a claim that a neural method improves tomography or a geological interpretation.

The classical objective for each training realization is a bounded, weighted, regularized least-squares problem in slowness perturbation q around the reference $s_0 = v_0^{-1}$:

$$\widehat{q}_\lambda = \arg\min_{q_{\min} \le q \le q_{\max}} \|W(Aq - y)\|_2^2 + \lambda \|Dq\|_2^2 + \eta \|q\|_2^2, \qquad v = (s_0 + q/1000)^{-1}. \tag{30}$$

The candidate λ is selected on a separate validation cohort; test families never choose it. The neural model sees measured residual features and ray coverage but no true velocity, family label or test score at inference. This comparison tests whether learned corrections help on shifted geometry, not whether they can memorize an acquisition.

The stored checkpoint is hash-checked before import and bounded against archive expansion. Request validation and resource bounds run before NumPy, SciPy or Torch are imported, so malformed large requests cannot trigger an expensive solver import. A user may submit 1 to 4096 rays; the domain and velocity bounds remain those of the fixed training system. A matching geometry and uncertainty flag only describes whether the supplied inputs match the training setup. It does not authorize a learned result when the new survey lies outside that domain, and it does not retrain or recalibrate the checkpoint.

The independent quadrature oracle samples each ray within cells and integrates slowness at a finer path discretization than the cell-length matrix. The difference between the training operator and this second forward calculation is reported as a discretization sensitivity. A well-fitting model under the training matrix can therefore still show a larger error under the independent integration. Both forward errors, the normalized residual, projected KKT measure, velocity bounds and model-to-reference metrics are retained in the artifact.

The negative conclusion is limited to these procedural families, noise law, 16×16 geometry and fixed comparison protocol. The two same-seed CUDA executions are not two independent samples from the training algorithm. They do, however, show that the observed failure is not reversed by the tested execution replay. A generalization claim would require

independent seeds, broader geologies and acquisition designs, contemporary matched baselines and eligible field data. The current user-input calculation is a straight-ray research instrument with explicit warnings, not a production neural velocity service.

Learned P and S phase picking

The phase-picking method implements a PhaseNet-family one-dimensional encoder-decoder for three classes: noise N, P arrival and S arrival. The product's PyTorch implementation is an original U-Net-style network, not a claim of bitwise equivalence to the official TensorFlow implementation. Each prepared example contains 6000 samples at 0.01 s spacing and three oriented components. The measured-input path validates source identity, component ordering, channel availability, sample interval and the trace-local normalization parameters before inference. Component baselines are removed over the declared initial window and each present component is divided by its peak absolute amplitude; missing components are represented by zeros with an explicit presence mask rather than inferred from other axes.

The encoder uses channel widths 16, 32, 64, 128 and 192, stride-two convolutions and residual blocks; the decoder upsamples and joins same-scale skip features. Seven-wide residual convolutions and group normalization preserve temporal context while controlling activation scale. Training targets are Gaussian probability curves centered on analyst annotations with a 15-sample spread. A weighted soft cross-entropy trains per-sample probabilities (p_N, p_P, p_S) whose values sum to one. Event and station identities define the disjoint split so near-duplicate traces from the same event or sensor cannot appear across training, validation and test partitions.

Inference is a thresholded pick selector, not a second trained model. For selected thresholds τ_P, τ_S and minimum time separation $\Delta_{\min}$, the selector is

$$\begin{aligned}
\widehat{t}_P &= \arg\max_t p_P(t) \quad \text{if } \max_t p_P(t) \geq \tau_P, \\
\widehat{t}_P &= \text{null} \qquad\qquad \text{otherwise}, \\
\widehat{t}_S &= \arg\max_t p_S(t) \quad \text{if } \max_t p_S(t) \geq \tau_S \text{ and } \arg\max_t p_S(t) - \widehat{t}_P \geq \Delta_{\min}, \\
\widehat{t}_S &= \text{null} \qquad\qquad \text{otherwise}.
\end{aligned} \tag{31}$$

At inference, analyst labels are not consulted. The documented selector uses a 0.4 s default P-S separation and can abstain independently for either phase. Evaluation compares every model on the same locked trace IDs and keeps misses, false picks, correct picks, out-of-tolerance picks and signed timing residuals separate. A threshold sensitivity grid is exploratory and is not used to retune the locked population benchmark.

The browser executes the exported ONNX model through WASM. The current parity receipt checks all 23 selected QC-valid held-out display traces against CPU ONNX arrays; the one QC-rejected trace remains excluded. The maximum same-checkpoint probability error was 4.8876×10^{-6}, with zero P/S peak displacement. This validates browser runtime parity on those records only: it is not independent PyTorch-oracle agreement, population accuracy, field transfer or phase truth. The curated lesson uses its own fixed 23-record display cohort and does not substitute it for the locked 6000-trace accuracy benchmark. The interface shows the waveform, class probabilities, analyst reference where available, thresholds, abstentions and timing errors on linked time axes. These views make comparison inspectable without treating a high probability as calibrated confidence. The phase-picking method remains unresolved for independent accuracy and full user-data interoperability even though the held-out browser asset parity check passes.

The training and evaluation pipeline begins from the source archive and records the exact waveform member identity, physical channel labels, analyst arrival metadata, event identifier and station identifier. Malformed archive members, duplicate identifiers, unsupported component sets and inconsistent sample windows are rejected before training. Splits are created from event and station groups with a recorded salt and partition digest, then frozen before model fitting. This prevents multiple windows from a single event or station from leaking across partitions. The STEAD-derived source is used under its documented attribution and license conditions; the product does not redistribute the entire source archive.

The target curve for class c at sample n, with annotated arrival n_c and target width $\sigma = 15$ samples, is

$$y_{c,n} = \exp\left[-\frac{(n-n_c)^2}{2\sigma^2}\right], \qquad \mathcal{L} = -\frac{1}{N}\sum_{i,n} w_{i,n} \sum_{c\in\{N,P,S\}} y_{i,c,n} \log \operatorname{softmax}(z_{i,n})_c. \tag{32}$$

The training code applies its declared class weighting and normalizes the Gaussian targets before the soft cross-entropy. The equation describes the supervised objective family; it does not imply that the probabilities are calibrated. The architecture's skip connections preserve sharp onset detail while the downsampled path aggregates longer context. A 60 s window is still finite context and can contain multiple events or emergent arrivals not represented by the target annotation.

The application displays probability curves in sample-aligned physical time, overlays reference picks only on evaluation records and reports the threshold used to accept or abstain. The P-S separation rule prevents an S pick from preceding or nearly coinciding with its P pick, but it is a simple selector rule and not a physical travel-time model. Browser runtime parity uses the same exported checkpoint and curated records, so it tests serialization and inference consistency rather than independent model quality. The complete phase-picker result must be judged by the locked population benchmark, group-disjoint split, per-phase metrics and analyst-reference quality together. None is replaced by the visual sharpness of a probability peak.

Geometry-normalized residual refinement

The separate geometry-normalized travel-time study asks whether a residual CNN supplied with two geometry-aware channels can improve on a validation-selected regularized straight-ray inverse across held-out geological families and acquisition layouts. It is a bounded synthetic comparison. It does not introduce null-space projection as new theory, and its failed result is retained rather than renamed as a successful method. The study uses a fixed 16 by 16 velocity grid, a reference model $v_{0,j}$, observed travel times y_i, and an explicit ray-cell length matrix A_{ij}. The sum of cell lengths on ray i is ℓ_i; c_j is total illumination of cell j. The normalized time residual and path-averaged cell feature are

$$\eta_i = \frac{y_i - (A\mathbf{s}_0)_i}{\ell_i}, \qquad c_j = \sum_i A_{ij}, \qquad h_j = \begin{cases} \frac{\sum_i A_{ij}\eta_i}{c_j}, & c_j > 0, \\ 0, & c_j = 0. \end{cases} \tag{33}$$

Here $\mathbf{s}_0 = 1/\mathbf{v}_0$ is reference slowness. Since y_i is in seconds and ℓ_i in metres, η_i has units s/m. The sum in the numerator of h_j has units seconds, and division by c_j gives s/m. The zero fill in unsupported cells is only a tensor-input convention; the coverage channel remains zero so the network can distinguish missing support from a measured zero residual.

The first channel linearizes velocity around the reference, using $\partial v/\partial s = -v_0^2$. The second records normalized support:

$$r_j = -\frac{v_{0,j}^2 h_j}{400\ \mathrm{m/s}}, \qquad g_j = \frac{c_j}{\max_k c_k}, \qquad \mathbf{x}_j = \begin{bmatrix} r_j \\ g_j \end{bmatrix}. \tag{34}$$

The tensor has shape $2 \times 16 \times 16$, ordered depth-major then horizontal. A common length-unit conversion cancels in g_j, but the feature is not invariant to changing the mesh: that alters ray-cell intersections, the coverage pattern, convolutional receptive field in metres, and the physical meaning of each pixel. Path normalization also does not whiten measurement noise. With independent time errors of variance σ_t^2, $\mathrm{Var}(\eta_i \mid A) = \sigma_t^2/\ell_i^2$; shorter paths have greater variance after this transformation. The network is not given the full covariance induced when paths share cells. The feature is therefore a deterministic back-projection, not an optimal preconditioner, exact inverse, adjoint gradient of the scoring operator, or posterior mean.

The residual network receives these two channels, applies a 3 by 3 convolution to 24 feature maps, three residual blocks, and a final one-channel projection. Its final layer starts at zero, making the initial prediction reproduce the reference velocity. A logit-space update constrains the prediction to the generator's declared 1400 to 4000 m/s interval. This bound is an authored synthetic range, not a field prior. Training uses 800 procedurally generated surveys, 160 validation surveys, Adam with learning rate 0.001, batch size 64, and 40 epochs. Checkpoint selection minimizes validation velocity RMSE. The objective combines normalized supervised velocity error with a forward travel-time term; truth labels are present only in training and evaluation, never as an inference input. The reference estimator is a validation-selected quadratic Tikhonov solve with first-difference regularization and a post-solve slowness clip. It is run on the same observations and evaluation cohorts.

The cell-length operator A constructs the features and training data term. An independently coded bilinear quadrature operator Q generates the synthetic observations and scores predicted travel times. This operator mismatch is disclosed: it does not establish field independence, and it prevents claiming that the feature and scoring path share one discretization. On the locked 160-case joint family and acquisition shift, learned-to-reference velocity RMSE is 2.194 and the independent travel-time RMSE ratio is 5.585; every case exceeds the validation-derived travel-time threshold. A second CUDA training with the same seed and manifests yields ratios 2.190 and 5.595. This is execution replay, not an independent-seed uncertainty estimate. Both runs fail the prespecified comparison. The evidence does not demonstrate improvement over modern eikonal, Bayesian, or learned tomography baselines because those systems were not reimplemented under matched operators and cohorts. Field validation, independent-seed training, broader geological generators, and a matched-operator ablation remain necessary before a different conclusion could be supported.

The comparison asks a narrow predictive question: does the learned correction improve the chosen reference under the fixed synthetic generator and the declared held-out split? It does not test whether the correction is geologically plausible when several velocity structures fit the same rays. The reference regularizer, velocity bounds, grid, and ray operator determine which alternatives are available to both methods. Changing the mesh changes cell coverage and convolutional scale, so a new grid would require a new training and validation protocol rather than a direct pixelwise transfer. The error ratios are dimensionless, but the underlying velocity and travel-time errors retain their physical units in the source receipts. Because every locked joint-shift case exceeds the selected threshold, aggregate success cannot be rescued by good in-distribution behavior. Future tests should predeclare a matched fast-marching or eikonal reference, repeat training across indepen-

dent seeds, and evaluate eligible field picks with uncertainty-aware errors.

Executable procedure summaries

These algorithms describe the executable decisions and output records. They preserve unsupported and failed cases rather than replacing them with inferred conventions or successful-looking defaults.

Algorithm 1 Validate and screen an EDI transfer-function record

Require: EDI bytes, explicit unit and time convention, required components, error policy
Ensure: Validated transfer tensor, masks, QC reasons, or rejection record

Parse original bytes and preserve source hash and station identity
Validate finite positive frequencies, component arrays, and declared order
Resolve native E/B units and phase convention from explicit inputs
Check rotation, variance, tipper, remote-reference, and channel metadata
if orientation or conventions are unknown or internally inconsistent **then**
    return quarantined input with reason codes
end if
Convert to canonical E/H units; rotate only when covariance propagation is defined
Apply component masks without filling unsupported values
Evaluate necessary one-dimensional tensor consistency conditions
if the necessary screen fails **then**
    retain the measured QC artifact; do not run layered one-dimensional inversion
else
    expose the validated transfer function to the selected layered-earth workflow
end if
Export source identity, normalized tensor, masks, units, decisions, and hashes

Algorithm 2 Select a bounded layered-earth MT model

Require: QC-passed complex impedance, fixed thicknesses, bounds, solver, and seed
Ensure: Re-evaluated final model, residuals, identifiability diagnostics, and status

Encode resistivity in log space and initialize from the declared 100 Ωm model
Form the complex-component data objective and adjacent log-resistivity penalty
Apply bounded TRF, projected Adam, or the per-sounding neural optimizer
for each accepted update **do**
    Recompute the objective and persist a copy of the accepted model state
end for
Re-evaluate the selected final model with the layered forward recursion
Calculate active residual, bounds, local linearized rank, and condition number
if the forward state or declared identifiability criteria fail **then**
    mark unresolved and preserve the reason codes
else if synthetic recovery beats the independent start under locked criteria **then**
    mark conditional synthetic recovery
else
    mark unresolved; data fit alone does not imply geological accuracy
end if
Export parameters, state identity, predictions, residuals, diagnostics, and provenance

Algorithm 3 Frequency-continuation acoustic FWI with receiver holdout

Require: Acoustic observations, initial model, fixed geometry, noise scale, and frequency stages
Ensure: Final velocity, active and withheld predictions, objective history, and verdict

```
Freeze receiver holdout before selection; exclude it from gradient computation
Estimate the declared coarse background using training receivers only
for each frequency stage from low to high do
    Configure the stated filter, finite-difference propagator, and model bounds
    for each optimization update within the fixed budget do
        Propagate each training shot and compute the stage objective
        Differentiate and apply the bounded L-BFGS model update
        Persist model, optimizer state, objective terms, and stopping code
    end for
end for
Recompute predictions at the exact selected final model for all receivers
Calculate active and withheld noise-normalized errors and model error when truth exists
if termination is invalid or either receiver split violates the locked criterion then
    mark unresolved or negative control with its cause
else if model recovery improves the independent start and both splits pass then
    mark synthetic recovery under the declared case only
else
    retain unresolved status; lower training loss is not validation
end if
Export arrays, units, split identity, model state, stage history, and source hashes
```

4.1 Potential-field inverses

The gravity and magnetic experiments use SimPEG rectangular-prism sensitivity operators [1, 2], explicit survey geometry, uncertainty-weighted data residuals, and spatial regularization. The canonical potential-field setup uses a 10,752-cell mesh and 256 surface observations (docs/research/scientific-implementation.md). L2 and smoothed-L1 IRLS solutions are distinct estimators, not interchangeable render styles. Magnetic cases include scalar induced assumptions and vector magnetization so that remanence can be exposed as model misspecification. Choclo prism values [8] provide an independent physical check for sign and scale in specified comparisons. Sensitivity and conditional noise ensembles are not interpreted as calibrated geological posterior intervals.

The current result table includes a gravity deep-body example for which weighted WRMS is 1, while model correlation is only 0.283 and recovered centroid depth is approximately 570 m versus a synthetic truth near 282 m. This is not a paradox: the inverse fits the observations at the chosen noise scale while regularization and limited information admit a different model. The structure is unresolved despite the weighted data fit.

4.2 MT, ERT, and waveform processing

The MT response is calculated through the complex impedance recursion for isotropic layers; differentiable physics-guided inversion is a related approach [3]. EDI ingestion is a transfer-function workflow and preserves conventions, units, rotations, masks, and stated variances; EDI is not treated as raw time-series data. Multiple optimizers (bounded least squares, Adam, and a small per-sounding neural parameterization) address matched ob-

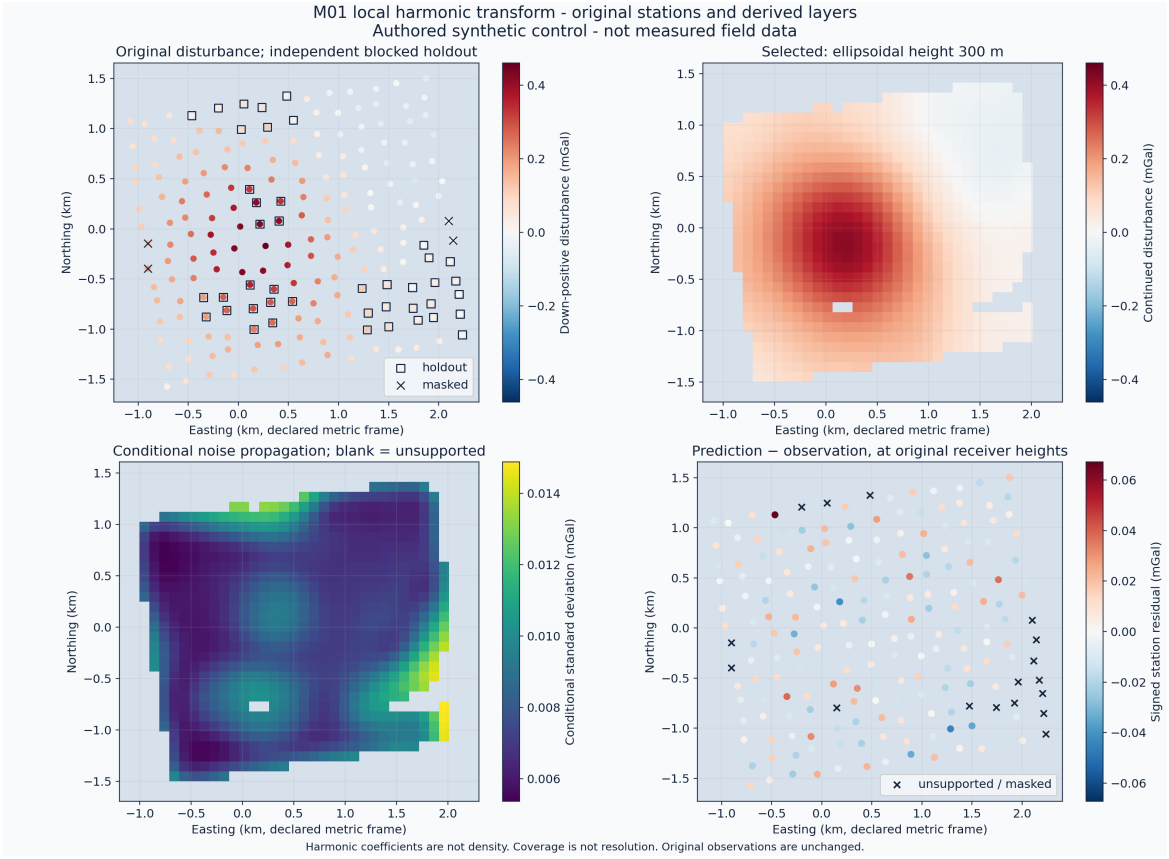


Figure 1: Authored gravity processing control. Upper left: original station disturbance and blocked holdout or masked locations. Upper right: harmonic continuation to 300 m. Lower left: propagated conditional noise standard deviation, with blank cells unsupported. Lower right: signed prediction residuals at original receiver locations. The figure is synthetic and does not represent field measurements.

jectives in synthetic fixtures. A neural architecture related conceptually to physics-guided inversion [3] is not asserted to reproduce the cited paper's architecture or published results.

ERT workflows validate electrode/ABMN structure, QC and topographic factors before inversion, following established 2D methods such as PyGIMLi [9]. The numerical examples distinguish authored controls from absent or restricted field assets and retain failed inversions. Seismic preprocessing distinguishes instrument counts, response-corrected units, filters, gaps, and sample windows. STA/LTA picking and learned phase inference consume explicitly specified traces; analyst picks are reference annotations, not subsurface truth.

4.3 Travel-time tomography, FWI, and joint inversion

Travel-time inversion uses specified ray geometry and a training/withheld partition. Its velocity maps are interpreted together with ray support; unsupported cells do not receive equivalent evidential status. The bilingual first-arrival lesson calls the product inverse core on an authored refracting two-layer survey and compares complete withheld shots with a training-only homogeneous baseline. This is a bounded synthetic calculation.

The FWI implementation uses constant-density acoustic propagation [4, 5], a fixed acquisition family, bounded velocity, a data-derived coarse background, and matched update budgets for full-band and frequency-continuation schedules. Active and withheld

receiver WRMS accompany model RMSE. A lower model RMSE can coexist with residuals above the specified noise level. The four-case computation includes a layered reflection, a fault, a salt/cycle-skipping negative control, and a low-velocity channel. Its case-level results are not generalized to field-scale FWI or elastic/attenuating physics.

Joint inversion compares uncoupled and coupled estimates against the same observations and independent baselines. Cross-gradient coupling promotes structural similarity, while a Gaussian-mixture prior adds a distinct petrophysical assumption [10, 11]. These mechanisms are reported separately. In the shared synthetic case, cross-gradient slightly improves susceptibility RMSE while worsening density RMSE relative to the optimized uncoupled solution; the mixture-prior method slightly improves density RMSE while slightly worsening susceptibility. Neither is a joint two-property recovery gain. In a deliberately conflicting case, the negative control rejects a general claim that structural or petrophysical coupling must help.

4.4 Learned methods and residual refinement

The original gravity CNN predicts column density rather than a depth-resolved volume, reflecting the non-identifiability of depth from one potential-field map. Learned inverse benchmarks also show sensitivity to the sampled geological families [12]. Its comparison with a spatial L2 baseline uses identical seeded noisy observations and a shared column target. The autoencoder's threshold is selected on a separate calibration set and tested on withheld geometries; its measured failure to detect the withheld family is retained. These models are compact baselines and have not been compared with a broader set of published inverse methods.

The separate geometry-normalized residual manuscript reports a geometry-normalized, coverage-conditioned residual CNN against a validation-selected weighted straight-ray reference. Its input has two channels (normalized residual and coverage); the network has 24 hidden feature maps, not 24 input channels. The fixed protocol uses seed 77213, 40 epochs, 800 training cases, 160 validation cases, and held-out identity, family, acquisition, and joint shifts. On the 160-case joint holdout, the primary checkpoint has velocity and independent travel-time RMSE ratios 2.194 and 5.585; the second CUDA execution, with the same seed and manifests, has ratios 2.190 and 5.595. All 160 joint cases exceed each run's validation-derived threshold. The paper therefore records a negative result for its specified generator and operators. Repeating the same seed measures execution sensitivity; it does not estimate between-seed training variance. The source-bound receipts, checkpoints, bootstrap calculation and current limitations are in manuscripts/geometry-normalized-traveltime-refinement-v001/.

The method uses normalized residuals and ray-coverage maps as inputs, then predicts a bounded correction on the fixed grid. This design can condition corrections on sampled ray support, but it cannot recover unsupported structure without a generator-dependent prior. The weighted-L2 reference is selected on validation data; the held-out comparison does not retune it. Ratios above one favor that reference, and bootstrap intervals quantify resampling within these cases only. The experiment has no independent-seed estimate, no matched fast-marching baseline, and no field observations. Feature construction and bilinear scoring use nonidentical operators, a known mismatch that narrows the result's interpretation. The complete paper reports the objective, training protocol, ablations, receipts, and per-case failure analysis.

An earlier residual-v3.1/null-space experiment and its analysis remain archived, but they are not the current geometry-normalized experiment. Their older cohort, checkpoint, scorer and operator evidence cannot be combined with the current results. The archive

is retained for provenance at docs/research/archive/m12-nullspace-evaluation-draft-2026-11-13/; the product no longer declares it as a separate proposed-method paper. This distinction prevents the old positive aggregate ratios from being read as results of the current method or as a beyond-state-of-the-art claim.

7 Experimental protocol

The original cross-method catalogue contains 20 authored geometries, six controlled variants per case, and 348 method result records. These counts describe a frozen artifact family, not 348 independent geological samples. The variants change specific acquisition or physical factors such as contrast, noise, coverage, or regularization. Their outcomes are reported within each method's own variables and criteria. A shared catalogue organizes retrieval; it does not justify pooling unlike metrics into an overall success rate. Additional current-source work adds separate field-intake, survey, user-input, and learned cohorts, each with its own provenance and acceptance boundary.

For learned travel-time estimation, the fixed training protocol uses 800 training and 160 validation surveys, 40 epochs, and predeclared family, acquisition, and joint-shift evaluation cohorts. The geometry-normalized paper preserves case identifiers and receipts; its second CUDA run uses the same seed and manifests, so it measures execution sensitivity and not training-seed uncertainty. The exploratory ablations use new test-cohort seeds and do not replace the locked comparison. The phase picker has a separate locked 6,000-trace benchmark against analyst annotations. A 23-trace browser parity subset is selected for runtime comparison and is not a substitute for that cohort. FWI results use four stated synthetic models, including a salt cycle-skipping negative control and an unresolved low-velocity channel; saved output is distinguished from a fresh solve. These designs answer bounded questions and are not interchangeable.

Before each inverse, data are divided according to the relevant independence unit: flight line, shot, geological family, station, or trace. Spatially adjacent samples must not leak across a blocked holdout when the purpose is to assess transfer to new acquisition support. Hyperparameters are selected on the designated validation data; final evaluation uses sealed test data only after the selection is fixed. Records preserve the source bytes, preprocessing, coordinate and unit conventions, model grid, random seeds, engine versions, objective, stopping state, and output hashes. A missing provider-linked dataset remains unavailable and does not become a synthetic substitute. The protocol therefore prioritizes traceable negative and unresolved cases over a pooled headline score. Full current-course replay, all external data permissions, independent training seeds, and representative field validation are not yet complete. The reported benchmark sizes and splits are method-specific, and larger catalogue totals do not change their independence unit. No held-out result is reweighted after inspecting a failure.

8 Results

5.1 Reference-condition outcomes

The reference-condition summary lists 122 recovered, 184 unresolved, 6 failed, and 36 negative-control method results (source: docs/validation/results.md). These are case-specific outcomes. They must not be summed into an overall accuracy or treated as success probabilities because the cases, units, target variables, and criteria differ. Failure and unresolved outcomes are first-class evidence in this project.

5.2 FWI case matrix

The bilingual case-matrix lesson reads four hash-verified full-run arrays (notebooks/en/06-seismology/05-m10-fwi-case-matrix.ipynb; Spanish counterpart). Layered reflection and fault are labeled recovered under the case-specific criteria; salt remains a cycle-skipping negative control; and the low-velocity channel remains unresolved. For the channel, model RMSE improves relative to the starting model (ratio 0.521 for the continuation result), but fitted/withheld WRMS are 2.071/2.213 against the specified unit noise scale. This contrast shows why improvement from a starting model is not sufficient. The notebook re-analyzes saved output rather than re-solving FWI, and an independent propagator comparison is needed to test sensitivity to using the same forward engine.

5.3 EDI screening and phase picker

The observed cl061 EDI tensor is screened against necessary 1D consistency conditions and fails them. It remains QC-only. It provides no 1D resistivity model for cl061 and makes no geological interpretation from that station. For the phase picker, the 2026-10-27 local browser parity computation reports all 23 valid selected display traces matching CPU ONNX Runtime: maximum absolute probability difference 5.0664×10^{-6}, no P/S peak displacement, and matching accepted/abstained picks within one sample. A twenty-fourth selected trace is QC-rejected and never inferred. The separate locked 6,000-trace benchmark, not this curated 23-trace subset, is the population-level performance record. Even that benchmark compares predictions with analyst annotations and should not be described as geological truth.

5.4 Novelty and null-space assessment

Null-space methods are established approaches for examining non-uniqueness in travel-time tomography and seismic inversion [17, 18, 19]. The archived product draft does not introduce null-space projection as new inverse theory. Its post-hoc projection experiment also does not provide a matched current checkpoint/operator comparison that would isolate projection as the cause of an improvement. The archived aggregate ratios and threshold diagnostics remain historical calculations for their stated frozen cohorts; they are not outcomes of the current geometry-normalized manuscript and do not establish superiority over published methods.

The current geometry-normalized residual study is likewise a bounded synthetic comparison, not a novel solver claim. It fails both pre-registered joint-shift error ratios for both same-seed CUDA runs. The scientifically supported contribution is therefore a reproducible negative comparison with explicit operators, cohort identities, per-case diagnostics and a disclosed replay discrepancy. A new method-performance claim would require an independent-seed training protocol, broader generators, matched contemporary baselines, exact operator controls and validation on eligible field observations. No such claim is made here.

The catalog-wide plots and generated tables below make the comparison population inspectable without collapsing physically different errors into one score. Figure 2 separates gravity model correlation from centroid-depth error, while Figure 3 shows how one fixed intrusion responds to individually named acquisition perturbations. Table 3 preserves the case and solver rows behind those summaries. A data residual near its assumed noise level does not identify the location or extent of the causative body.

For layered MT, Figure 4 pairs active complex-component fit with log-resistivity model

error, and Table 4 retains solver status and starting-model error. Figure 5 and Table 5 show active and receiver-held-out waveform misfit for the FWI reference cases. Receiver holdout evaluates spatial transfer within the selected synthetic acquisition, not transfer to a new geology or field survey.

The joint-inversion panel in Figure 6 reports changes relative to optimized uncoupled property estimates; Table 6 includes both physical data residuals and evaluation status. Figure 7 and Table 7 compare learned column targets with the matched classical estimate, without interpreting anomaly flags as calibrated geology. Figure 8 and Table 8 report the frozen STEAD held-out benchmark across channel and signal perturbations; annotation uncertainty and benchmark population bound the interpretation.

Figure 9 counts method-by-variant verdict records by physical family. It is an inventory of outcomes, not a pooled accuracy rate, because the family denominators differ. Table 2 states case, variant, and method-key counts. Figure 10 and Table 9 expose the pre-registered geometry-normalized ratios with bootstrap intervals. For the joint-shift cohort, the intervals remain above the unit pass boundary, supporting the negative result rather than a performance claim.

Table 2: Committed experiment coverage by physical family, including case, variant, and method-key counts for this frozen release artifact.

Family	Cases	Variants	Method keys
gravity	4	24	2
joint	2	12	5
learned	2	12	4
magnetics	4	24	3
mt	4	24	3
seismic	4	24	2

Table 3: Gravity reference recovery reports data fit and geometric error separately for every case and solver.

Case	Method	WRMS	Correlation	RMSE ratio	Depth error (m)
GRAVITY_INTRUSION	l2	1	0.5753	0.8145	63.683
GRAVITY_INTRUSION	irls	1	0.5836	0.8106	38.472
GRAVITY_DEEP_BODY	l2	1	0.3567	0.7938	277.79
GRAVITY_DEEP_BODY	irls	1	0.3056	0.8092	282.76
GRAVITY_NOISY	l2	1	0.5067	0.8753	92.496
GRAVITY_NOISY	irls	1	0.5263	0.8682	84.091
GRAVITY_TILTED	l2	1	0.1578	0.8904	166.53
GRAVITY_TILTED	irls	1	0.1352	0.8962	183.28

Table 4: Layered MT reference results report active fit, initial-model error, and the explicit solver verdict on each synthetic sounding.

Case	Solver	Active WRMS	Initial WRMS	log-rho RMSE	Status
MT_RESISTIVE	mt-lm	1.022	24.375	0.17323	recovered
MT_RESISTIVE	mt-adam	1.022	24.375	0.1732	recovered
MT_RESISTIVE	mt-neural	1.023	24.375	0.21529	recovered
MT_CONDUCTIVE	mt-lm	0.9615	27.086	0.098159	recovered
MT_CONDUCTIVE	mt-adam	0.9615	27.086	0.098159	recovered
MT_CONDUCTIVE	mt-neural	0.9615	27.086	0.098159	recovered
MT_MIXED	mt-lm	0.9139	21.439	0.92457	unresolved
MT_MIXED	mt-adam	1.011	21.439	0.4252	recovered
MT_MIXED	mt-neural	0.9386	21.439	0.040128	unresolved
MT_NOISY	mt-lm	0.9034	19.208	0.031593	recovered
MT_NOISY	mt-adam	0.9034	19.208	0.030791	recovered
MT_NOISY	mt-neural	0.9034	19.208	0.031593	recovered

Table 5: Acoustic FWI reference cases preserve active fit, held-out fit, model ratios, and unresolved outcomes for each frozen model.

Case	Schedule	Model ratio	Active WRMS	Withheld WRMS	Status
FWI_LAYERED	fwi-l2	0.5347	1.13	1.146	recovered
FWI_LAYERED	fwi-multiscale	0.5269	1.208	1.233	recovered
FWI_FAULT	fwi-l2	0.7007	1.37	1.419	recovered
FWI_FAULT	fwi-multiscale	0.6533	1.34	1.388	recovered
FWI_CYCLE_SKIP	fwi-l2	0.8522	1.734	1.945	negative-control
FWI_CYCLE_SKIP	fwi-multiscale	0.8026	1.751	1.966	negative-control
FWI_NOISY	fwi-l2	0.6603	2.765	2.904	unresolved
FWI_NOISY	fwi-multiscale	0.5212	2.071	2.213	unresolved

Table 6: Joint priors are compared with optimized uncoupled estimates while both physical residuals remain visible for each structure.

Case	Method	Density ratio	Magnetic ratio	Gravity WRMS	Magnetic WRMS	Status
JOINT_SHARED	joint-uncoupled	1	1	1	1	unresolved
JOINT_SHARED	joint	1.002	0.9991	1.039	1.004	unresolved
JOINT_SHARED	pgi	0.9947	1	1.017	1.023	unresolved
JOINT_CONFLICT	joint-uncoupled	1	1	1	1	recovered
JOINT_CONFLICT	joint	1.001	0.9989	1.009	1.014	negative-control
JOINT_CONFLICT	pgi	1	1	1	1	negative-control

Table 7: Learned gravity outputs compare column targets with classical estimates; flags are not calibrated geological classifications.

Case	Method	Column RMSE	Classical RMSE	Ratio	AE flag	Status
LEARNED_CNN	l2	75.71	–	–	–	unresolved
LEARNED_CNN	cnn	80.1	75.71	1.058	–	unresolved
LEARNED_CNN	autoencoder	–	–	–	False	unresolved
LEARNED_AUTOENCODER	l2	84.9	–	–	–	unresolved
LEARNED_AUTOENCODER	cnn	90.99	84.9	1.072	–	unresolved
LEARNED_AUTOENCODER	autoencoder	–	–	–	False	unresolved

9 Validation

Validation is staged so that one successful gate cannot stand in for a different claim. Parser tests check byte bounds, structure, coordinate and unit declarations, masks, and rejection

Table 8: Frozen STEAD held-out phase scores use analyst annotations, a fixed half-second arrival tolerance, and named signal perturbations.

Condition	Model	Phase	Precision	Recall	F1	Median abs. error (s)
nominal	M00	P	0.142	0.168	0.154	1
nominal	M00	S	0.0739	0.0876	0.0802	2.99
nominal	M08	P	0.0771	0.0908	0.0834	4.8
nominal	M08	S	0.0233	0.0272	0.0251	7.27
nominal	M13	P	0.981	0.981	0.981	0.03
nominal	M13	S	0.959	0.952	0.955	0.05
drop_E	M00	P	0.142	0.168	0.154	1
drop_E	M00	S	0.0739	0.0876	0.0802	2.99
drop_E	M08	P	0.0461	0.0544	0.0499	4.81
drop_E	M08	S	0.0126	0.0148	0.0136	7.92
drop_E	M13	P	0.983	0.979	0.981	0.03
drop_E	M13	S	0.938	0.926	0.932	0.07
drop_N	M00	P	0.142	0.168	0.154	1
drop_N	M00	S	0.0739	0.0876	0.0802	2.99
drop_N	M08	P	0.0452	0.0534	0.049	4.81
drop_N	M08	S	0.0121	0.0142	0.013	7.86
drop_N	M13	P	0.98	0.979	0.979	0.03
drop_N	M13	S	0.936	0.927	0.931	0.07
drop_Z	M00	P	0.142	0.168	0.154	1
drop_Z	M00	S	0.0739	0.0876	0.0802	2.99
drop_Z	M08	P	0.0515	0.0608	0.0558	4.81
drop_Z	M08	S	0.0216	0.0254	0.0233	7.68
drop_Z	M13	P	0.912	0.893	0.903	0.07
drop_Z	M13	S	0.952	0.919	0.935	0.06
noise_0p1	M00	P	0.142	0.168	0.154	1
noise_0p1	M00	S	0.0739	0.0876	0.0802	2.99
noise_0p1	M08	P	0.513	0.593	0.55	0.27
noise_0p1	M08	S	0.562	0.535	0.548	0.24
noise_0p1	M13	P	0.924	0.653	0.765	0.05
noise_0p1	M13	S	0.954	0.613	0.746	0.08
shift_plus_800	M00	P	0	0	0	8
shift_plus_800	M00	S	0	0	0	9.13
shift_plus_800	M08	P	0.0469	0.0552	0.0507	12.1
shift_plus_800	M08	S	0.0101	0.0118	0.0109	15.9
shift_plus_800	M13	P	0.98	0.981	0.981	0.03
shift_plus_800	M13	S	0.959	0.952	0.956	0.05

Table 9: Preregistered refinement cohorts report candidate-to-baseline error ratios and bootstrap intervals; values above one favor the baseline.

Shift cohort	n	Velocity ratio	95% CI	Time ratio	95% CI	Flags
id test	160	3.034	[2.942, 3.124]	4.957	[4.748, 5.172]	14
family only	80	2.178	[2.097, 2.275]	5.733	[5.626, 5.84]	80
acquisition only	80	3.092	[2.961, 3.229]	5.268	[4.987, 5.576]	6
joint ood	160	2.194	[2.133, 2.264]	5.585	[5.509, 5.662]	160

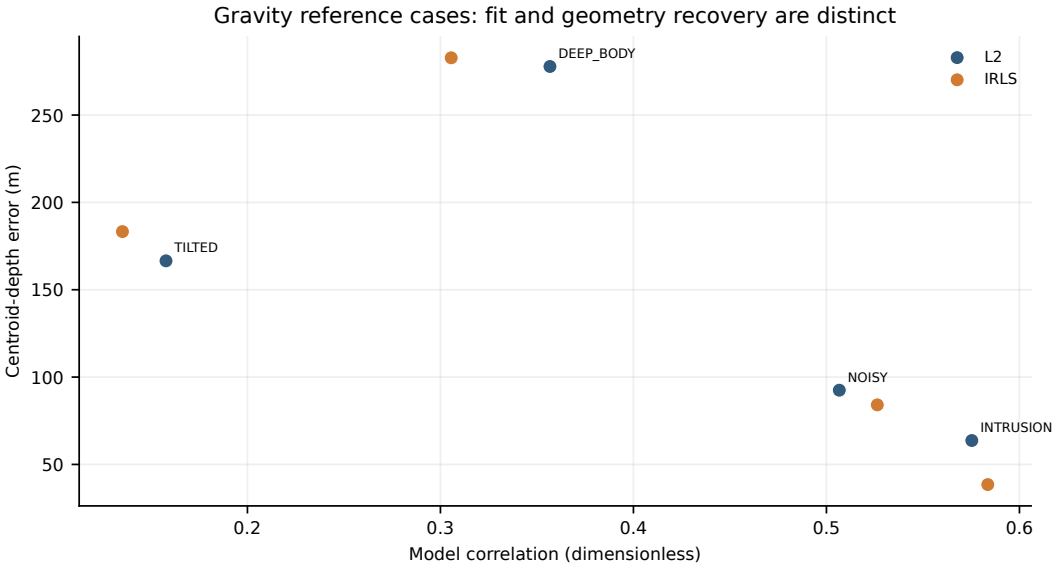


Figure 2: Reference gravity cases separate density-model correlation from centroid-depth error across the frozen experiment catalog of synthetic bodies.

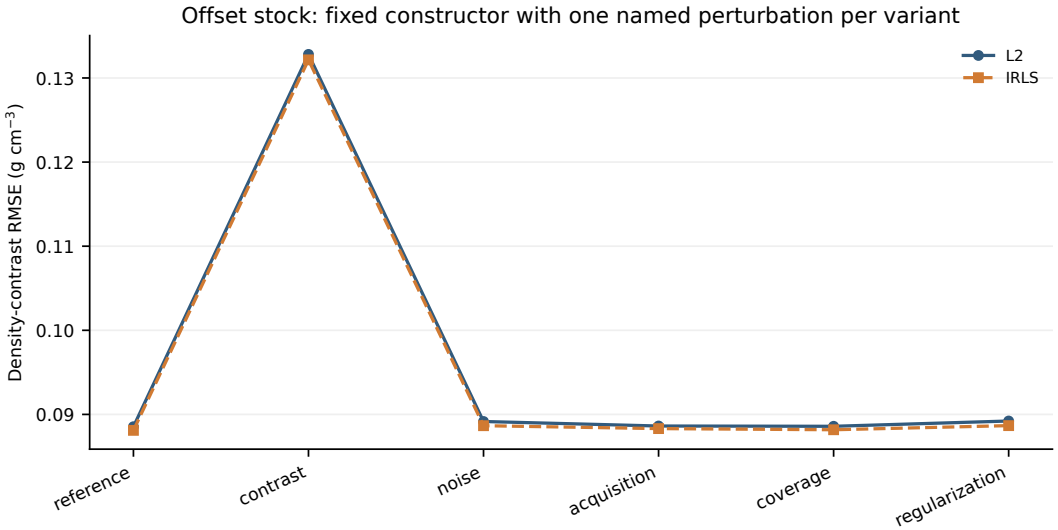


Figure 3: A single gravity intrusion case shows model error under individually named acquisition perturbations with its forward and inverse settings fixed.

reasons. Forward tests compare analytical or library identities on small controlled problems; for selected prism calculations, an independent physical implementation checks signs and magnitudes. These tests can expose implementation errors, but an oracle comparison covers only the geometries and precision tested. Gradient checks compare a local derivative with finite differences and do not establish a global optimum or convexity.

Inverse validation separately records solver termination, data residual, model error where truth exists, prior sensitivity, spatial coverage, and held-out predictions. The gravity example demonstrates that a noise-scaled data fit may coexist with low model correlation and a centroid-depth error. A low-velocity FWI case improves model RMSE over its starting model but remains unresolved because fitted and withheld residuals exceed the specified noise scale. The learned travel-time refinement fails both predeclared joint-shift ratios in two same-seed executions, with every test case above its run-specific

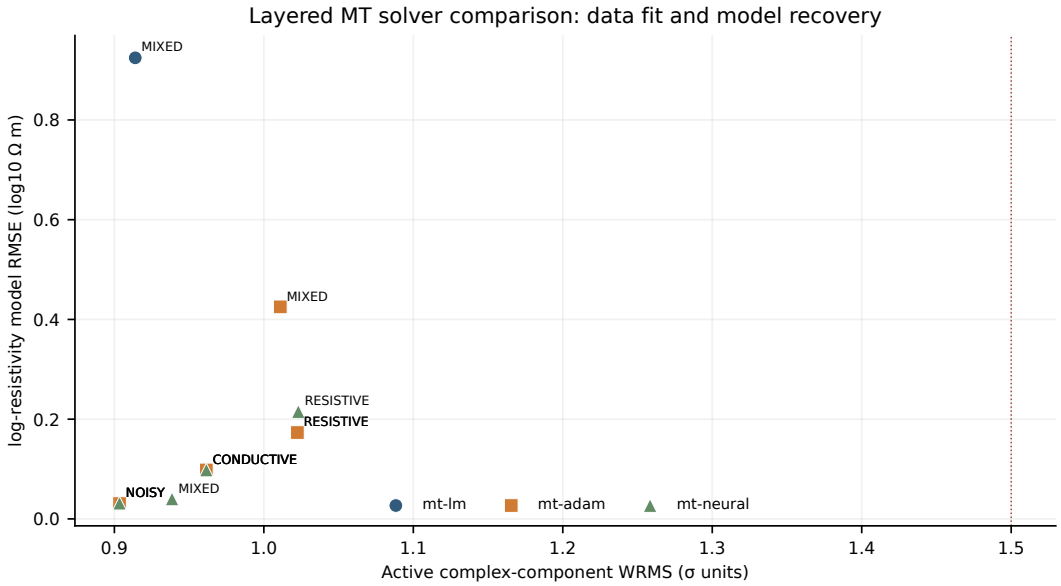


Figure 4: Layered MT reference cases compare active complex-response fit with resistivity-model error across the catalogued solver implementations.

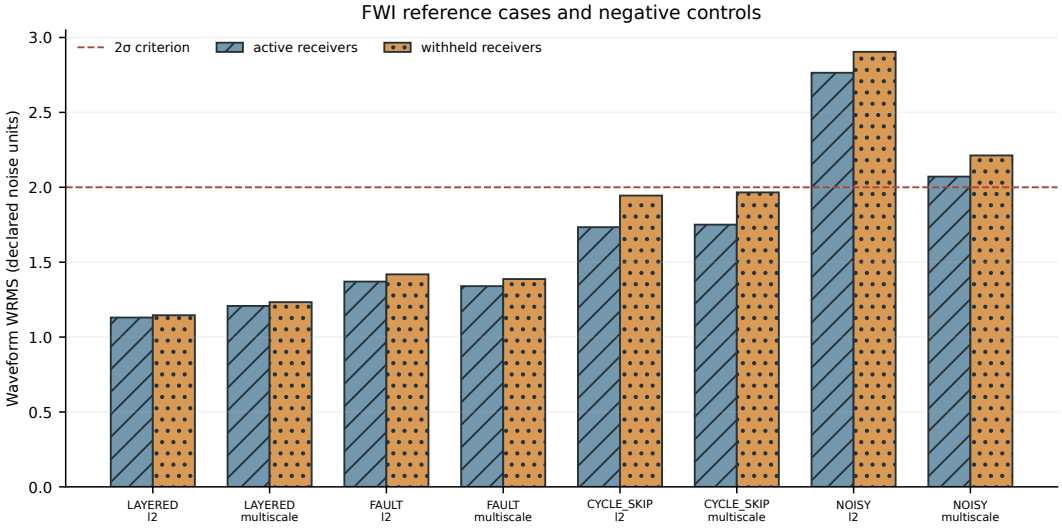


Figure 5: Acoustic FWI reference cases retain active and receiver-held-out waveform residuals beside the declared noise-scale criterion.

validation threshold. Those values remain negative evidence, not a reason to move the threshold after seeing the test results. The measured Clear Lake cl061 transfer function fails the necessary 1D consistency screen and is not passed to a layered inversion.

Reproducibility checks bind sources, arrays, environments, seeds, and manifests; same-checkpoint CPU/WebAssembly parity checks only runtime consistency on the selected traces. Independent-oracle parity, cohort accuracy, field transfer, and user acceptance are separate tests. The technical report currently passes LaTeX compilation and the method matrix is structurally complete, but the manuscript-depth checker still fails its required figures, tables, references, algorithms, and coverage fields. Notebook depth and the SDD convergence ledger also remain incomplete. Consequently, none of these local checks constitutes whole-product scientific acceptance or release approval. This section states the evidence layers and retains their failure states rather than combining

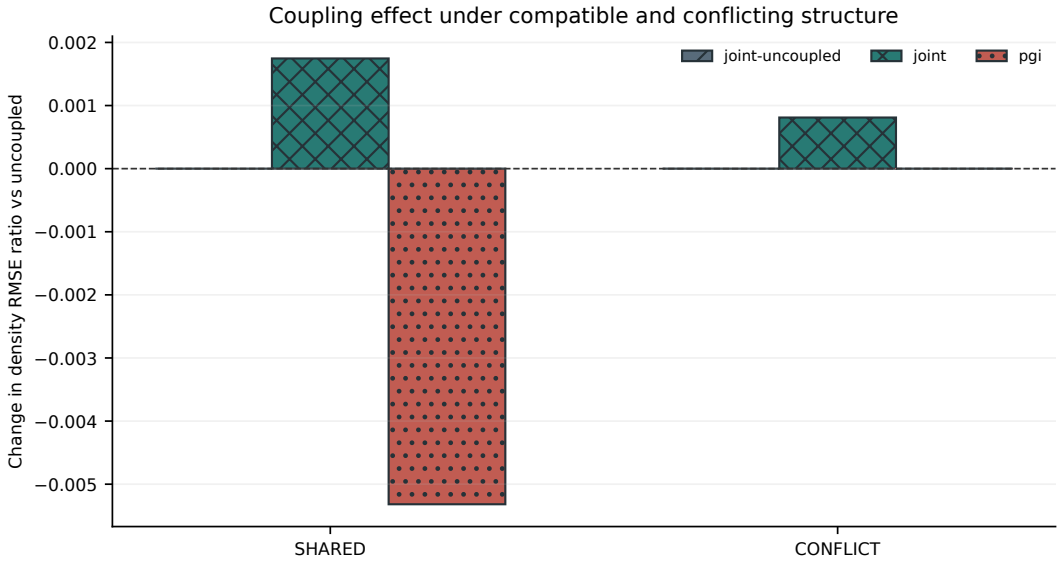


Figure 6: Joint inversion controls show density-error changes relative to optimized uncoupled estimates for compatible and conflicting structures.

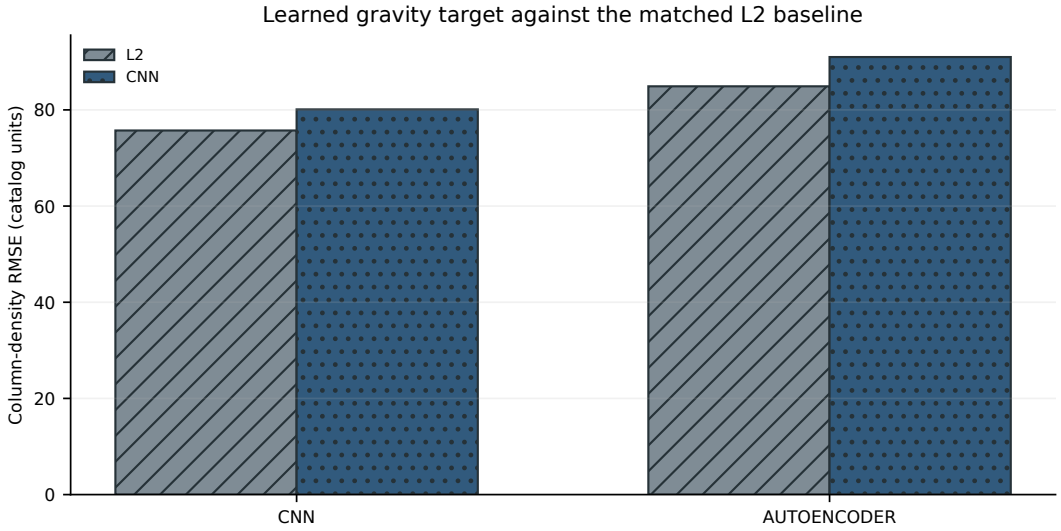


Figure 7: Learned gravity column estimates and matched L2 baselines are compared on the same catalogued observations and targets.

them into a single validation score. The data-derived summary tables are regenerated by `scripts/build_report_assets.py` from the committed experiment catalog, frozen phase benchmark, and geometry-normalized analysis receipt. The script refuses an incomplete catalog and records method metrics at the case level before aggregation. This provenance check protects the displayed numbers but is not an independent implementation of every underlying solver. The plots preserve the comparison denominators, and the tables retain terminal status so that unresolved rows remain discoverable during scientific review.

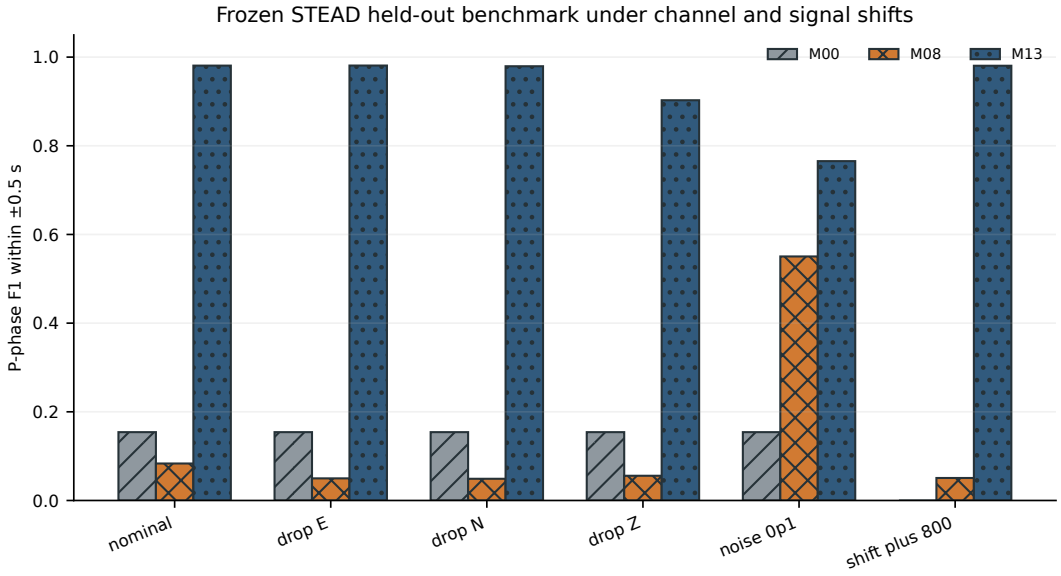


Figure 8: Frozen STEAD held-out P-phase F1 scores compare three pickers under channel removal, added noise, and a fixed timing shift.

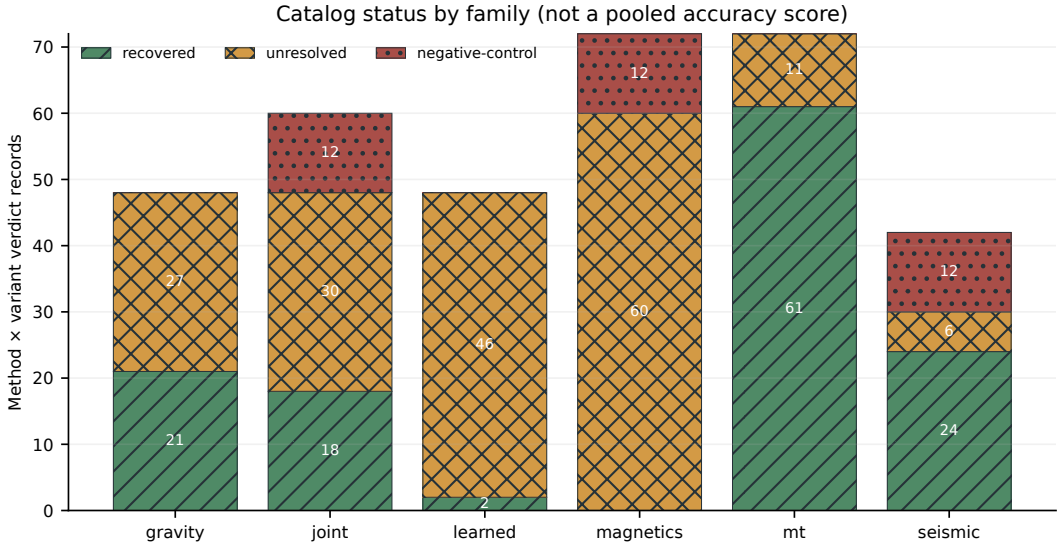


Figure 9: Catalog verdict counts retain recovered, unresolved, and negative-control method variants by physical family without pooling accuracy.

10 Discussion

The bilingual notebook sequence covers inverse-problem foundations, gravity, magnetics, MT, ERT, seismology, joint inversion, and learning. Lessons include progressively deeper content: from units and forward physics to actual solver calls, held-out evaluation, sensitivity analyses, independent oracles, and negative controls. The notebooks are intended as research and teaching materials, but no educational efficacy study has been performed. The source course is cited as inspiration; its instructional materials are not copied into the product.

Scientific visualization is organized around the method's actual state variables. Potential-field scenes show geometry, cells, survey observations, cuts, and physical property scales;

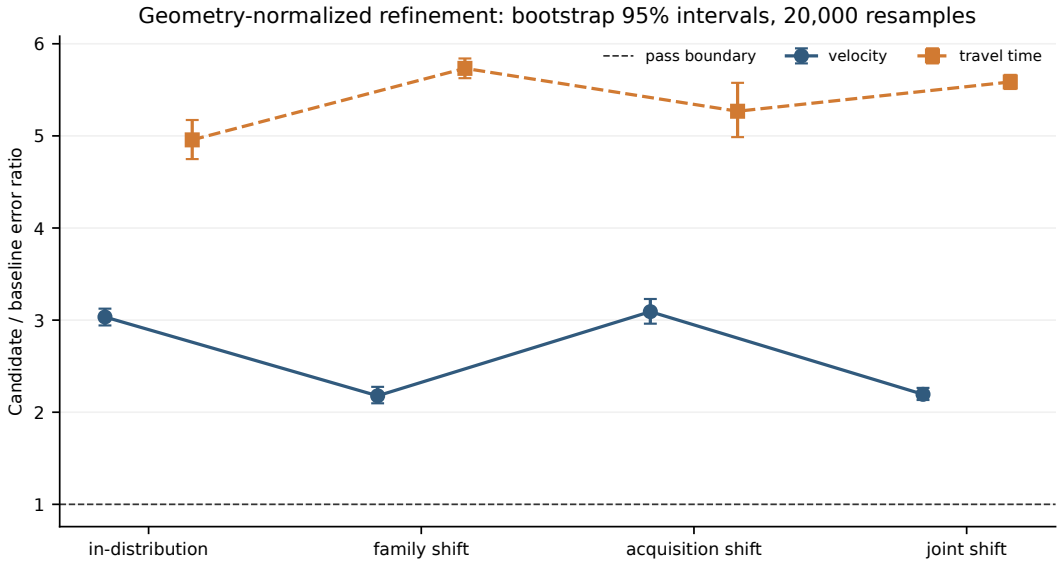


Figure 10: Geometry-normalized refinement error ratios and bootstrap intervals are shown against the preregistered unit pass boundary.

MT views show layer structure and frequency response; seismic instruments relate wavefields, receiver gathers, and inversion state; learned methods show their actual target arrays and comparison metrics. Camera rotation changes viewpoint, not physics. Time cursors and optimizer-state replay address different quantities and are labeled accordingly. A rendered state alone is not evidence of uniqueness, resolution, or uncertainty.

The scientific views expose physical geometry, survey support, and computed states. Their effect on interpretation has not been measured in a user study, and browser performance has not been characterized across devices, routes, languages, and themes.

The strongest methodological conclusion is that observation fit, parameter recovery, geological interpretation, and transfer to new data are separate questions. Potential-field inverses can fit within a noise model while leaving large structural uncertainty; coupled objectives can trade one property against another; learned models can perform well on their training family and fail under a combined shift. The report's examples are chosen to expose these distinctions with explicit targets and negative controls. They do not imply that the catalogue spans the variability of real geology, or that one selected case can represent a basin or survey population.

The software contribution is consequently methodological infrastructure: it brings data custody, numerical operators, solver settings, evaluation criteria, and presentation into a traceable workflow. That infrastructure can make it easier to inspect assumptions, but it does not by itself create novel inverse theory. The separate geometry-normalized study tests a specific learned correction and fails its prespecified shifted cohort. Its negative result is useful evidence about that feature, network, generator, and comparator. The experiment does not demonstrate that the method is beyond the state of the art, because published contemporary methods were not reproduced under matched operators and data splits. For the geometry-normalized experiment, model and travel-time ratios use the same held-out cases but different outputs and therefore need not have equal scale. Bootstrap intervals describe resampling variation within the frozen cohort; they do not represent variation across training seeds, geology generators, or survey regions. The independent operator mismatch between feature construction and the bilinear scoring operator remains a threat to construct validity, even though it is disclosed and replayed. These

limits prevent the current figures from supporting an algorithmic superiority claim.

The implementation should be read as a research environment with unevenly mature verticals. A local parser or inverse call can be real without making the entire user-data workflow complete. Public visitors can inspect documented cases and run supported browser calculations; protected project flows and heavy server jobs still require secure host-side qualification. The current VPS release remains the legacy version, so the candidate's UI, user-data paths, and authenticated jobs have not been accepted live. This operational distinction matters when assessing claims made from local artifacts or screenshots.

The work therefore leaves concrete questions open: how much structure is identifiable under each acquisition, how sensitive is the estimate to the chosen prior and grid, which data-quality failures should block an inverse, and do the linked views improve expert review? Answering them requires matched field data, independent solvers, calibrated uncertainty, and measured user studies. The present report supplies definitions and executable examples for asking those questions; it does not report those studies as complete.

11 Computational reproducibility

All 60 bilingual notebooks were run from top to bottom in fresh kernels on 30 October 2026. The computation comprised 57 unchanged notebooks and three refreshed notebooks; none ended with an execution error. Focused fresh-kernel replays of the refreshed examples then passed. The complete replay used `scripts/run_notebooks.py --check` with external scratch storage. Notebook sources are under `notebooks/en/` and `notebooks/es/`.

Numerical outcomes are tied to their method-specific computation and date. The FWI case matrix reads hash-verified outputs and does not rerun the expensive inverse; its source is notebooks/en/06-seismology/05-m10-fwi-case-matrix.ipynb and the Spanish counterpart. The learned-velocity notebook computation dated 28 October and browser-parity computation dated 27 October are documented computations in docs/validation/m12-checkpoint-notebook-2026-10-28.md and docs/validation/m13-browser-population-parity-2026-10-27.md. The null-space refinement results and cohort definitions are in docs/design/features/m12-velocity-validation/nullspace-v1_1-protocol.md. Reference-condition metrics are in docs/validation/results.md. Method equations, implementation details, and primary-source mapping are in docs/research/scientific-implementation.md, docs/research/review.md, and docs/design/SDD.md.

Notebook replay verifies execution for the recorded source and environment; it is not an independent physical oracle. Scientific replication requires using the method-specific data, environment, split, seed, operator, and comparison criteria together. A lesson that reads an archived computation must not be described as having recomputed that solve.

12 Limitations and open research questions

1. **Field validation and eligibility.** Most quantitative recovery results are synthetic. Provider data with incomplete datum, uncertainty, rights, or provenance remain unresolved or ineligible. cl061 is a QC failure for 1D inversion, not a field inverse.
2. **Non-uniqueness and uncertainty.** Conditional ensembles do not sample geological posterior uncertainty. Potential-field structure remains prior- and parameterization-dependent. Coverage and sensitivity do not by themselves yield calibrated resolution.

3. **FWI scope.** The experiments use bounded acoustic physics and fixed synthetic acquisition families. Source uncertainty, realistic correlated noise, anisotropy, elasticity, attenuation, broader cycle-skipping regimes, and an independent propagator remain necessary research directions.
4. **Joint inversion.** Cross-gradient and mixture-prior outcomes depend on their scale, prior, and structural compatibility. The limited calibration matrix does not establish general superiority over optimized single-modality estimates.
5. **Machine learning.** The learned-velocity calculation fails its pre-specified joint-shift comparison. The positive null-space variant is narrow synthetic evidence and needs external reproduction and broader cohorts. The phase-picker's 23-trace parity set is not an accuracy cohort; the separate 6,000-trace benchmark uses analyst annotations whose uncertainty limits the interpretation of timing scores.
6. **Data and reproducibility.** Some numerical tests require retained external inputs, including joint-inversion matrices and magnetic controls. Missing inputs are not replaced with synthetic stand-ins. Rights-limited source files are not copied merely to make a computation executable.
7. **Service capacity.** Local solver tests, builds, and browser parity do not establish worker admission, VPS capacity, cancellation, or recovery. Replacement-service capacity and operational behavior remain unmeasured.
8. **Educational effectiveness.** A bilingual executable notebook collection is a learning resource, but no controlled study establishes learning outcomes or comparative pedagogical effectiveness.

Useful next studies include preregistered geological-family and acquisition/noise ensembles; matched contemporary travel-time baselines; independent-seed training and calibration under forward-model and geological uncertainty; independent FWI propagation; licensed field inversions with source/datum/error review; and user studies of whether linking assumptions, predictions, residuals, and uncertainty improves interpretation. The scope of external datasets is also bounded by licensing, redistribution rights, and metadata quality. A public catalog entry does not imply that all source bytes are stored locally. Likewise, an upload accepted by a parser is not automatically eligible for every inverse: a method may require a coordinate reference, error model, acquisition design, or dimensionality condition that the file does not establish. Explicit refusal and quarantine states are therefore part of the scientific boundary, alongside successful calculations. Until the user workflow is qualified end to end, local solver coverage should not be read as a claim that each visitor can process every survey.

13 Conclusions

Inverse Earth Studio combines source-linked geophysical processing and inverse methods with explicit assumptions, independent physical checks where available, and preserved negative evidence. Its central examples show that data fit, model recovery, transfer, and visual plausibility are different questions. The workbench includes synthetic and bounded real-data screening calculations, bilingual executable notebooks, and computational paths across multiple physical methods.

These computations do not establish field validity or generality across geological settings. Several method-specific comparisons, end-to-end user-supplied-data workflows, actual-host measurements, independent solver comparisons, and user-facing visual evaluations remain to be completed. The current learned travel-time refinement fails its two joint-shift error criteria in the primary run and a same-seed replay. The earlier null-space draft

is archived as a historical analysis of an established operation, not as a novel method. The work provides a reproducible software and experimental framework that makes assumptions and counterexamples inspectable. Algorithmic novelty and publication priority are not established. The quantitative tables and graphics deliberately preserve negative and unresolved outcomes alongside successful checks. They document what the implemented cases establish today and make the remaining evidence requirements concrete: eligible field data, matched independent baselines, calibrated uncertainty, and qualified end-to-end user workflows. Those requirements are prerequisites for stronger scientific and product claims; they are not implied by the breadth of the method catalogue.

14 Data and code availability

Source code is available at https://github.com/fsantibanezleal/CAOS_Geophysics. Scientific methods, notebooks, test data, computational records, and source-rights decisions are linked above and organized in `data/`, `data-pipeline/`, `docs/`, `notebooks/`, `tests/`, and `models/`. Some external raw observations are absent because redistribution rights or field eligibility remain unresolved. The intended application origin is https://geophysics.ml.fasl-work.com/. Local computations do not establish replacement-service capacity. The data-derived report assets are rebuilt locally with `.venv-pipeline/Scripts/python.exe scripts/build_report_assets.py`; the report is compiled from `manuscripts/inverse-earth-studio-report-0.04.003/main.tex.in` using the repository's LaTeX build instructions. The frozen catalog is at data/derived/v2/catalog.json, held-out phase scores are at data/derived/phase/stead-heldout-benchmark.json, and the separate refinement analysis is at manuscripts/geometry-normalized-traveltime-refinement-v001/generated/analysis.json. Their inclusion does not grant redistribution rights to omitted source observations or convert synthetic cohorts into field evidence. The repository retains code, hashes, configurations, and derived outputs needed to inspect the reported values, subject to source licenses and the documented data-availability decisions.

References

[1] R. Cockett, L. J. Heagy, A. Pidlisecky, and D. W. Oldenburg, "SimPEG: An open source framework for simulation and gradient based parameter estimation in geophysical applications," *Computers & Geosciences*, vol. 85, pp. 142–154, 2015. doi:10.1016/j.cageo.2015.09.015.

[2] L. J. Heagy, R. Cockett, S. Kang, G. K. Rosenkjaer, and D. W. Oldenburg, "A framework for simulation and inversion in electromagnetics," *Computers & Geosciences*, vol. 107, pp. 1–19, 2017. doi:10.1016/j.cageo.2017.06.018.

[3] P. Goyes-Peñafiel et al., "Physically guided deep unsupervised inversion for 1D magnetotelluric models," *IEEE Geoscience and Remote Sensing Letters*, 2025. doi:10.1109/LGRS.2025.3528767.

[4] J. Virieux and S. Operto, "An overview of full-waveform inversion in exploration geophysics," *Geophysics*, vol. 74, no. 6, pp. WCC1–WCC26, 2009. doi:10.1190/1.3238367.

[5] Deepwave, "Full waveform inversion example and documentation." https://ausargeo.com/deepwave/example_fwi.

[6] S. M. Mousavi, Y. Sheng, W. Zhu, and G. C. Beroza, "STanford EArthquake Dataset (STEAD): A global data set of seismic signals for AI," *IEEE Access*, vol. 7, pp. 179464–179476, 2019. doi:10.1109/ACCESS.2019.2947848.

[7] W. Zhu and G. C. Beroza, "PhaseNet: A deep-neural-network-based seismic arrival-time picking method," *Geophysical Journal International*, vol. 216, no. 1, pp. 261–273, 2019. doi:10.1093/gji/ggy423.

[8] Fatiando a Terra, *Choclo: geophysical forward modelling with equivalent sources*. https://www.fatiando.org/choclo/latest/overview.html.

[9] PyGIMLi, *Inversion guide and ERT examples*. https://www.pygimli.org/user-guide/inversion/.

[10] SimPEG, *Joint inversion with cross-gradient regularization*. https://docs.simpeg.xyz/latest/content/user-guide/tutorials/13_joint_inversion/plot_inv_3_cross_gradient_pf.html.

[11] T. Astic et al., "Joint inversion of geophysical data with a petrophysical prior." https://arxiv.org/abs/2002.09515.

[12] W. Lu et al., "OpenFWI: Large-scale multi-structural benchmark datasets for seismic full waveform inversion." https://github.com/lanl/OpenFWI.

[13] A. G. Mantilla, *Curso teórico-práctico sobre inversión geofísica en Python*, source commit `0084021fcc6c1d2f112b144e6e5f0a3d92f20dff`. https://github.com/Anagabrielamantilla/inversion-geofisica-python.

[14] MTpy v2, *Magnetotelluric data and EDI documentation*. https://mtpy-v2.readthedocs.io/en/stable/index.html.

[15] U.S. Geological Survey, *Magnetotelluric data, Clear Lake region, northern California*. doi:10.5066/P14KAQ3M.

[16] EarthScope, *EMTF transfer-function collection: Clear Lake*. doi:10.17611/DP/EMTF/GMEG/Clearlake.

[17] R. W. L. de Wit, J. Trampert, and R. D. van der Hilst, "Toward quantifying uncertainty in travel time tomography using the null-space shuttle," *Journal of Geophysical Research: Solid Earth*, vol. 117, B03301, 2012. doi:10.1029/2011JB008754.

[18] Y. Liu et al., "Square-root variable metric-based nullspace shuttle: A characterization of the nonuniqueness in elastic full-waveform inversion," *Journal of Geophysical Research: Solid Earth*, 2020. doi:10.1029/2019JB018687.

[19] A. Torres and M. D. Sacchi, "Deep decomposition learning for reflectivity inversion," *Geophysical Prospecting*, 2023. doi:10.1111/1365-2478.13366.

[20] Y. Li and D. W. Oldenburg, *3-D inversion of magnetic data*, *Geophysics*, vol. 61, no. 2, pp. 394–408, 1996. doi:10.1190/1.1443968.

[21] Y. Li and D. W. Oldenburg, *3-D inversion of gravity data*, *Geophysics*, vol. 63, no. 1, pp. 109–119, 1998. doi:10.1190/1.1444302.

[22] T. G. Caldwell, H. M. Bibby, and C. Brown, *The magnetotelluric phase tensor*, *Geophysical Journal International*, vol. 158, no. 2, pp. 457–469, 2004. doi:10.1111/j.1365-246X.2004.02281.x.

[23] J. R. Peacock, K. Kappler, L. Heagy, T. Ronan, A. Kelbert, and A. Frassetto, *MTH5: An archive and exchangeable data format for magnetotelluric time series data*, *Computers & Geosciences*, vol. 162, 105102, 2022. doi:10.1016/j.cageo.2022.105102.

[24] G. Göktürkler, *Seismic first-arrival tomography with functional description of traveltimes*, *Journal of Geophysics and Engineering*, vol. 6, no. 4, pp. 374–385, 2009. doi:10.1088/1742-2132/6/4/005.

[25] C. Bunks, F. M. Saleck, S. Zaleski, and G. Chavent, *Multiscale seismic waveform inversion*, *Geophysics*, vol. 60, no. 5, pp. 1457–1473, 1995. doi:10.1190/1.1443880.

[26] E. Haber and D. W. Oldenburg, *Joint inversion: a structural approach*, *Inverse Problems*, vol. 13, no. 1, pp. 63–77, 1997. doi:10.1088/0266-5611/13/1/006.

[27] C. Rücker, T. Günther, and F. M. Wagner, *pyGIMLi: An open-source library for modelling and inversion in geophysics*, *Computers & Geosciences*, vol. 109, pp. 106–123, 2017. doi:10.1016/j.cageo.2017.07.011.

[28] Y. Wu and Y. Lin, *InversionNet: An efficient and accurate data-driven full waveform inversion*, *IEEE Transactions on Computational Imaging*, vol. 6, pp. 419–433, 2020. doi:10.1109/TCI.2019.2956866.

[29] J. Sun, K. Innanen, T. Zhang, and D. Trad, *Implicit seismic full waveform inversion with deep neural representation*, *Journal of Geophysical Research: Solid Earth*, vol. 128, no. 3, e2022JB025964, 2023. doi:10.1029/2022JB025964.

[30] H. Lan and Z. Zhang, *Topography-dependent eikonal equation and its solver for calculating first-arrival traveltimes with an irregular surface*, *Geophysical Journal International*, vol. 193, no. 2, pp. 1010–1026, 2013. doi:10.1093/gji/ggt036.

[31] N. Rawlinson and M. Sambridge, *Seismic traveltime tomography of the crust and lithosphere*, *Advances in Geophysics*, vol. 46, pp. 81–199, 2003. doi:10.1016/S0065-2687(03)46002-0.

[32] J. A. Sethian and A. M. Popovici, *3-D traveltime computation using the fast marching method*, *Geophysics*, vol. 64, no. 2, pp. 516–523, 1999. doi:10.1190/1.1444558.